

Name: _____

A2 Physics Unit 3

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /100	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	Max UMS
2017	43	50	57	64	71	85
2018	38	46	54	63	72	90
2019	34	41	49	57	65	81
2022	17	26	36	46	57	85
2023	26	36	46	56	66	82

Surname	Centre Number	Candidate Number
Other Names		2



GCE A LEVEL – NEW

1420U30-1



PHYSICS – A2 unit 3
Oscillations and Nuclei

THURSDAY, 15 JUNE 2017 – MORNING

2 hours 15 minutes

For Examiner's use only			
	Question	Maximum Mark	Mark Awarded
Section A	1.	11	
	2.	10	
	3.	15	
	4.	13	
	5.	18	
	6.	13	
Section B	7.	20	
Total		100	

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space use the continuation page(s) at the back of the booklet taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

This paper is in 2 sections, **A** and **B**.
Section **A**: 80 marks. Answer **all** questions. You are advised to spend about 1 hour 35 minutes on this section.
Section **B**: 20 marks; Comprehension. You are advised to spend about 40 minutes on this section.
The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **4(a)**.

SECTION A

Answer all questions.

1. (a) Describe an experiment that uses various absorbers to identify the type(s) of radiations emitted by a radioactive sample. Justify all steps in the experiment. [4]

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- (b) A radioactive sample of material has a half-life of 11.4 days and an initial activity of A_0 . Determine:

- (i) the decay constant; [2]

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(ii) the activity of the sample after 57.0 days in terms of A_0 ; [2]

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(iii) the **percentage decrease** in the number of nuclei in the sample after 57.0 days. [3]

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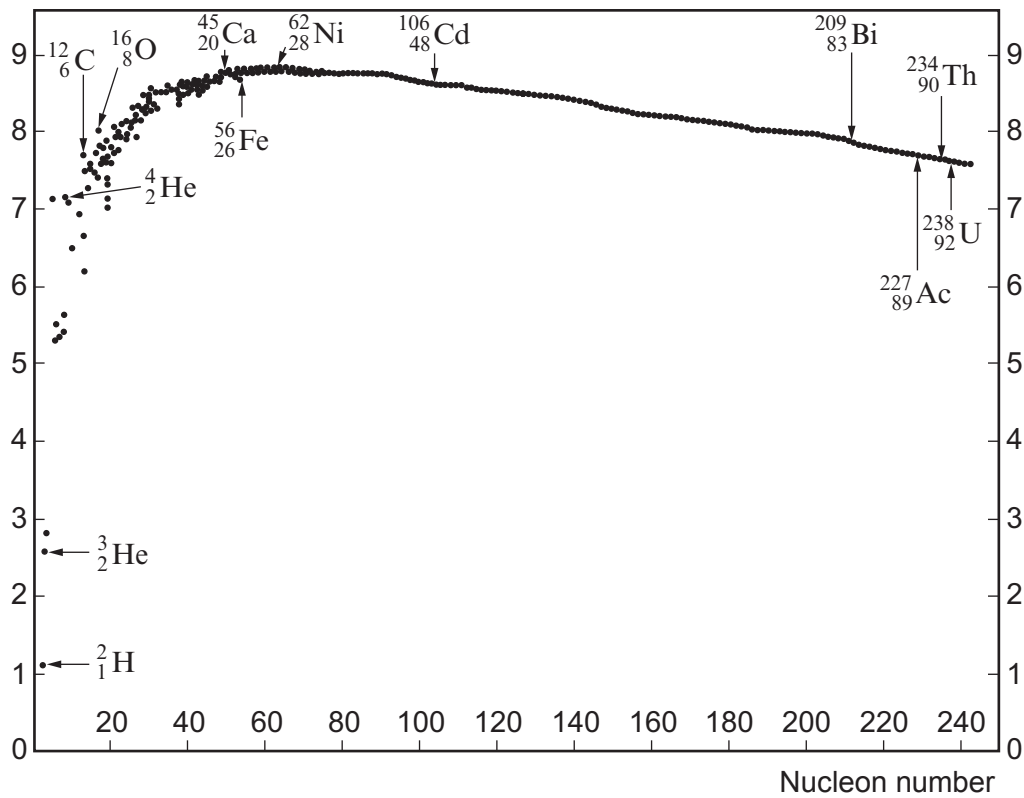
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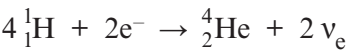
2. (a) Nuclear energy can be released by fusion or fission. Describe these processes by considering the graph of the binding energy per nucleon plotted against the nucleon number. [4]

Binding energy per nucleon
(MeV/nucleon)



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(b) The fusion of hydrogen to helium in the Sun may be represented by:



Particle	Mass / u
${}^1_1\text{H}$	1.00728
${}^4_2\text{He}$	4.00151
e^-	0.00055
ν_{e}	0.00000

$1\text{ u} = 931\text{ MeV}$

Calculate the energy released in this reaction in MeV. [3]

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(c) Nuclear power has played a role in the generation of electricity in Wales. Wylfa Newydd on Anglesey may feature in further developments. Indicate a benefit and an issue that may arise from such projects and discuss their relative importance. [3]

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3. (a) The first law of thermodynamics is given by:

$$\Delta U = Q - W$$

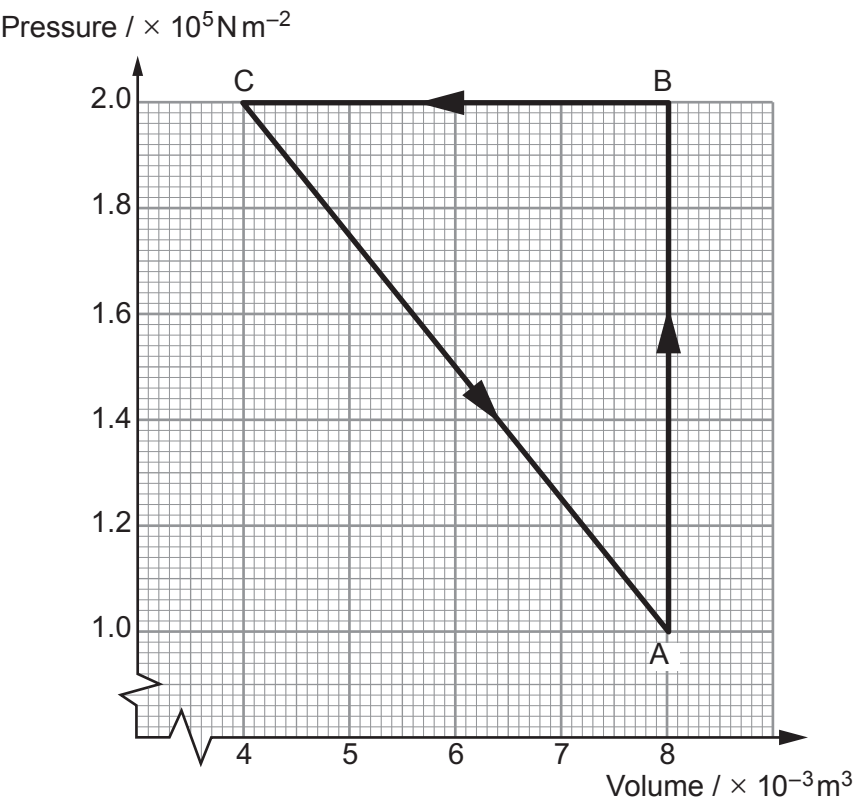
State what is represented by: [3]

ΔU

Q

W

(b) A fixed mass of gas is taken around the closed cycle $A \rightarrow B \rightarrow C \rightarrow A$.



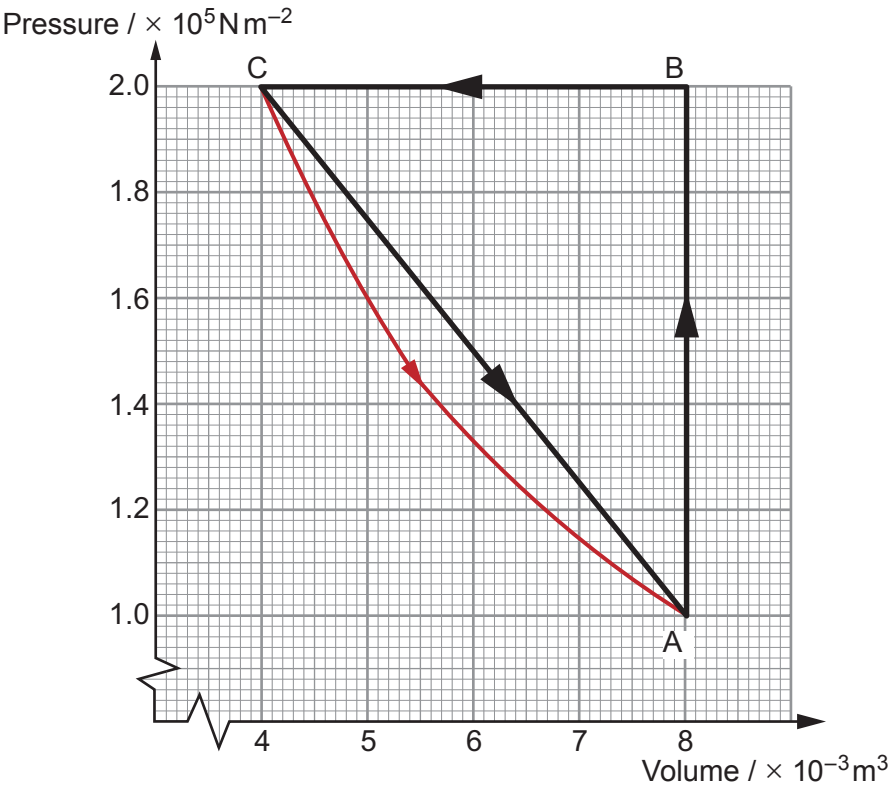
(i) Complete the table by describing each process in terms of pressure and volume. The description for process $A \rightarrow B$ is already inserted. For each process state if any work is done, and if so indicate if it is done *on* or *by* the gas. *No calculations are required.* [5]

Process	Description of process	Work done on / by gas (if any)
$A \rightarrow B$	Increase in pressure at constant volume	
$B \rightarrow C$		
$C \rightarrow A$		

(ii) Determine the net work done on the gas during the entire cycle.

[2] Examiner only

(c) The same mass of gas is taken around the cycle $A \rightarrow B \rightarrow C \rightarrow A$ for a second time. For this cycle the temperature is kept constant between C and A. The new path from C to A is shown in red on the graph below, together with the original path.



(i) Show that the temperature is constant along the new path from C to A.

[3]

(ii) Estimate the difference in the net work done in the two cycles.

[2]

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4. (a) Heat is supplied to an ideal gas in a sealed container of fixed volume. Explain carefully the physical processes that cause an increase in the **temperature and pressure** of the gas. [6 QER]

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- (b) A container of fixed volume contains oxygen gas at a temperature of 293 K.
- (i) Five oxygen molecules have speeds 400, 425, 450, 550 and 625 m s⁻¹. Determine their rms speed. [2]

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- (ii) Using an appropriate calculation, determine whether or not the rms speed calculated in part (b)(i) is consistent with the expected rms speed of the molecules of the gas at this temperature. (Relative molecular mass of oxygen gas = 32.) [3]

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- (iii) If the gas in the container is heated and the pressure of the gas increases by 20% of its initial value, determine the rms speed of the molecules of the gas. [2]

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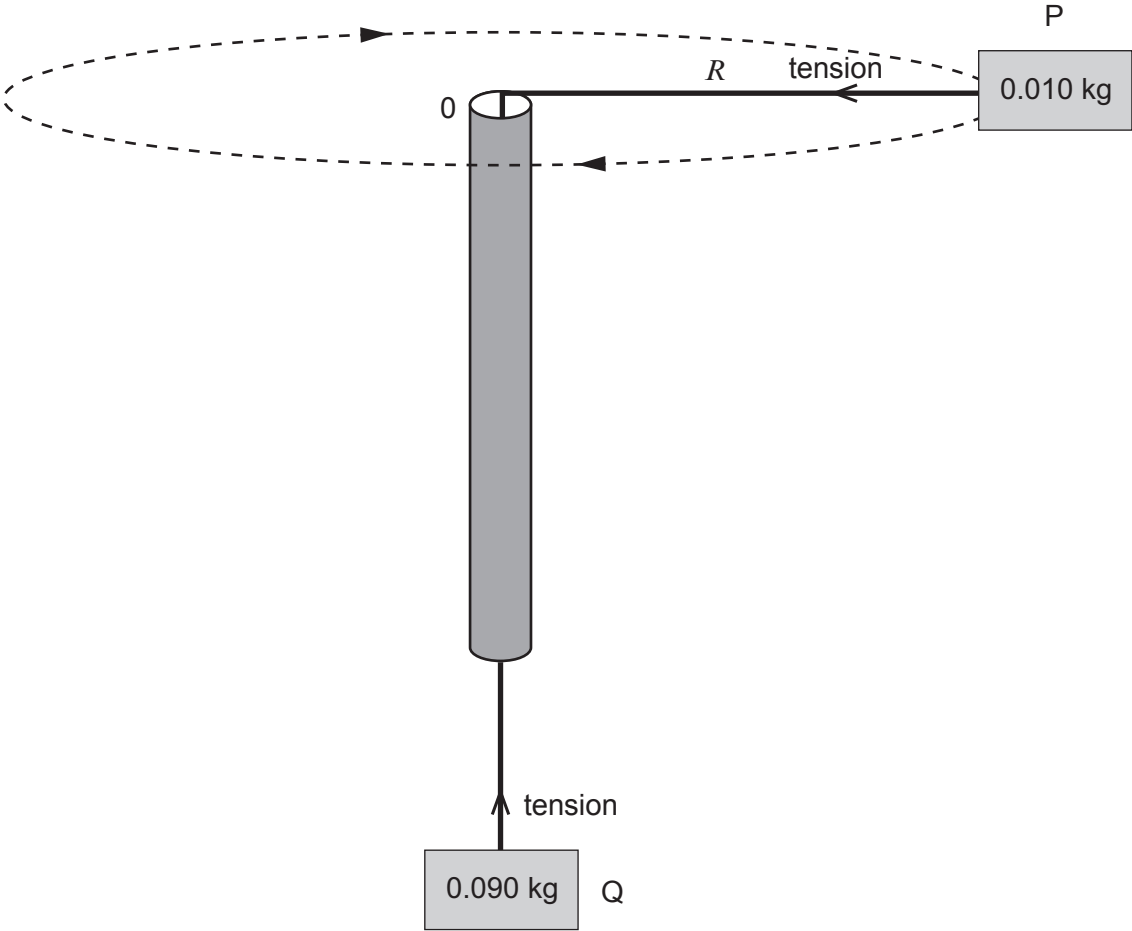
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5. A piece of string is threaded through a hollow narrow cylinder. Two small objects, P and Q, with masses 0.010 kg and 0.090 kg respectively are attached to the ends of the string, as shown.

A student holds the cylinder and sets the 0.010 kg mass rotating in a horizontal circle of radius R , which is kept constant at 0.50 m. The time for 10 rotations is recorded. The tension in the string provides both the centripetal force on P and an upward force to hold Q in equilibrium.

The measurement is repeated for different values of R . All measurements are recorded in the table overleaf.



- (a) Show that the speed, v , of mass P for each measurement is given by:

$v = \frac{2\pi R}{T}$ where T is the period of rotation. [1]

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(b) Complete the table.

[4]

R / m	Time for 10 rotations / s	Period T / s	v / ms^{-1}	$v^2 / \text{m}^2 \text{s}^{-2}$
0.50	4.7			
0.60	5.2			
0.70	5.6			
0.80	6.0			
0.90	6.3			

(c) (i) Assuming that OP is horizontal, write an equation relating the centripetal force to v and R . [2]

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(ii) Hence, by using the equation for the forces acting on mass Q, show that:

$$v^2 = 9g R$$

where g is the acceleration due to gravity.

[3]

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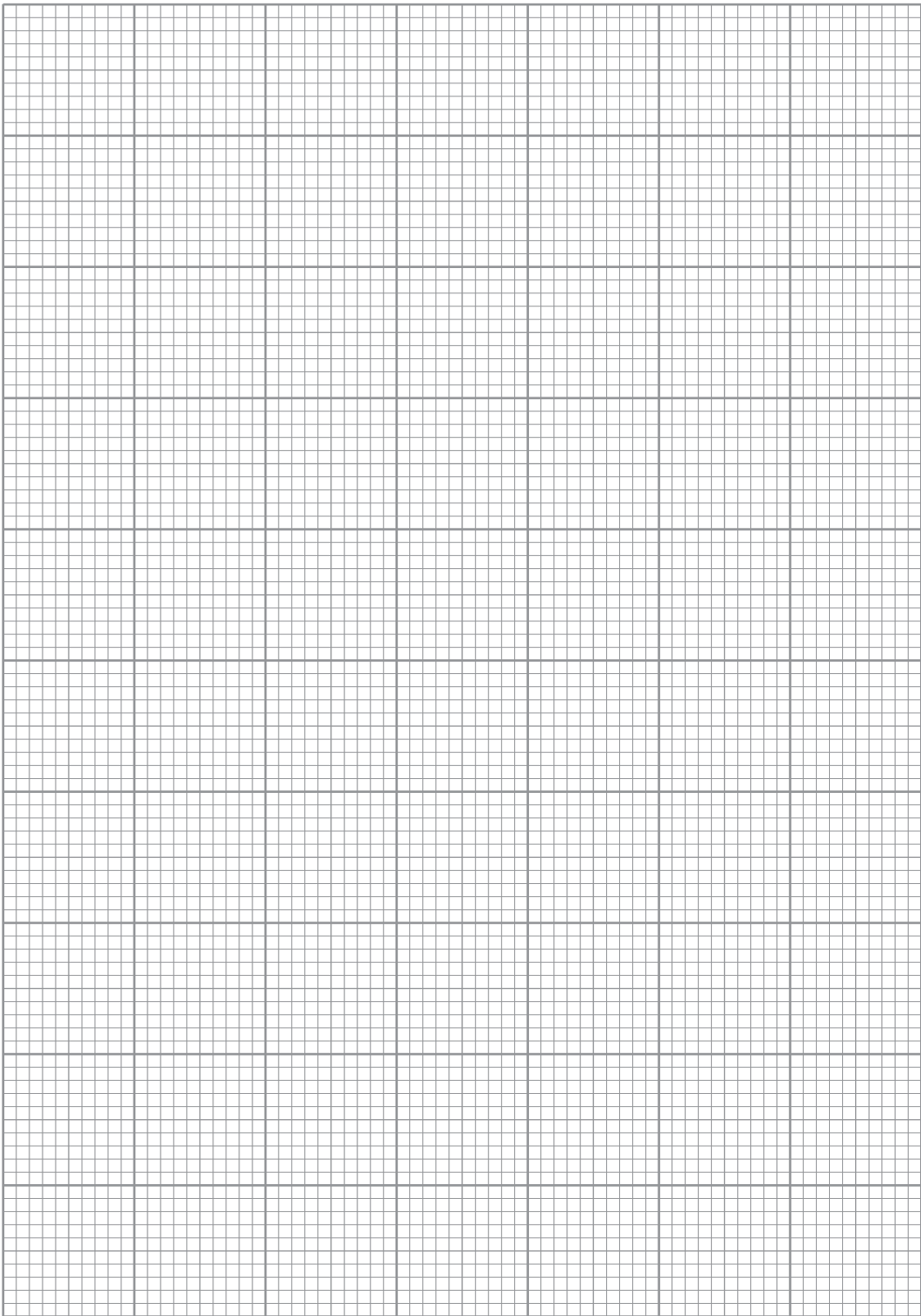
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(d) (i) Use the data in the table to plot a graph of v^2 (y -axis) against R (x -axis).

[4]

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(ii) Determine a value for g .

[3]

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(iii) Suggest a way in which the experiment can be improved.

[1]

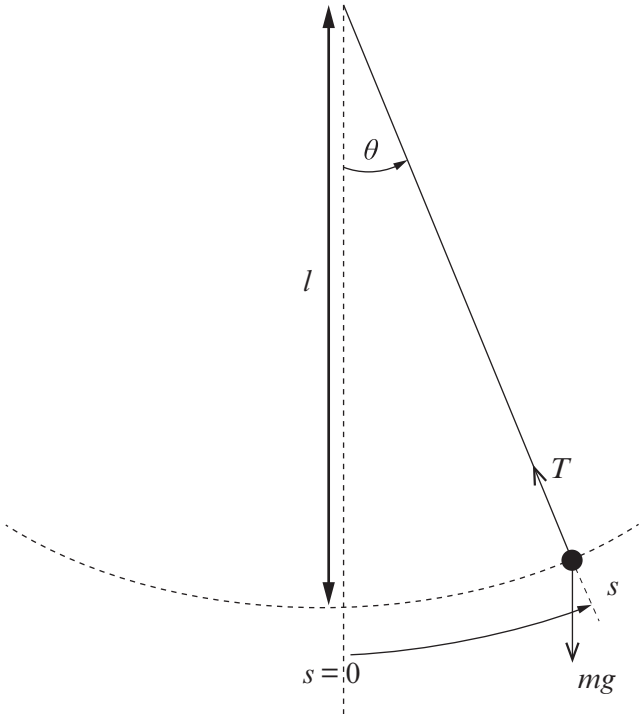
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6. (a) The pendulum in the figure below has a small bob of mass, m , suspended at the bottom of a light string of length, l . The string is shown at an angle, θ , to the vertical and the mass, m , is at a distance, s , along the arc. The forces acting on the mass are shown.



- (i) Name the **two** forces acting on the mass. [1]

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- (ii) By considering these forces show that:
resultant force component on the mass along the arc = $-mg \sin \theta$

You may add to the diagram if you wish. [3]

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- (iii) If the oscillation is small so that $\sin \theta \simeq \theta$ show in clear steps that the acceleration along the arc may be written as: [2]

acceleration = $-\frac{gs}{l}$

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- (iv) Discuss whether the equation in part (a)(iii) satisfies the definition of simple harmonic motion. [2]

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(b) A small mass oscillates at the lower end of a pendulum of length 1.20 m.

- (i) Determine its:

I. period; [2]

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II. frequency. [1]

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- (ii) If the maximum displacement angle of the mass is 0.067 rad, justify the use of simple harmonic motion in part (b)(i). [2]

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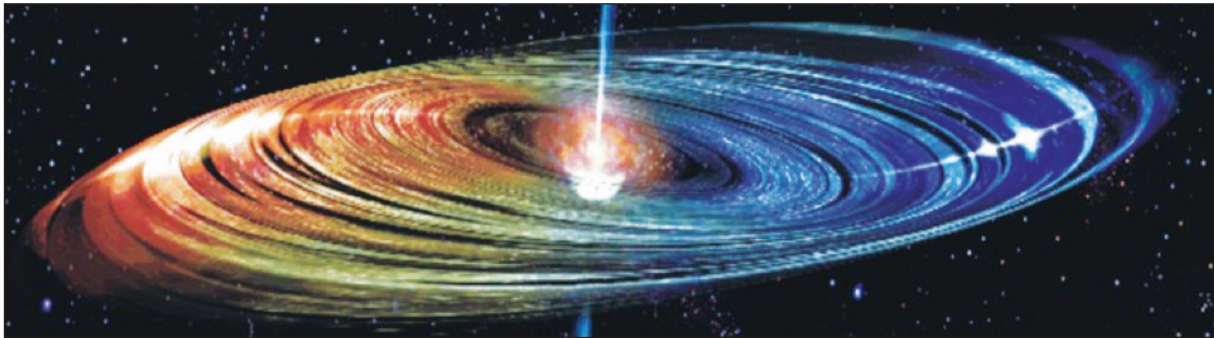
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SECTION B

Answer all questions.

Read through the following article carefully.

Lasers in Outer Space	Paragraph
Einstein’s theory of stimulated emission, published in 1916, laid the foundation for lasers but it took humans another 44 years before the first successful ruby laser was produced. However, seven years beforehand a maser had been produced. This is similar to a laser but involves microwaves instead of light – microwave amplification by stimulated emission of radiation.	1
It turns out that masers are easier to build than lasers because the lifetime of metastable energy levels tends to be proportional to $\frac{1}{\text{frequency of radiation}}$.	2
The ingenuity of humankind is often marvelled upon when such complicated devices as lasers and masers are used in devices such as DVD players and atomic clocks. However, nature itself is somewhat more modest – natural masers have been produced for billions of years in the atmospheres of stars, comets, star-forming regions, supernova remnants and even super massive black holes. And all this time the Universe has kept quiet about its technological achievements.	3
Essentially what happens in star masers is this – light and explosions from stars excite nearby gas regions. In these nearby gases, light and collisions lead to high energy levels becoming populated by electrons. Some of these higher energy levels will be metastable, setting up a maser amplifying region. As in any laser, the process has to start with a spontaneous emission accidentally shooting off in the correct direction, but afterwards, stimulated emission takes over and the maser beam starts towards infinity; infinity is stretching the truth a bit but it does sound good. In reality the intensity of the maser beam increases rapidly and the beam propagates at the speed of light as em radiation tends to do.	4
	
Diagram 1	
In all fairness to humans, there is one aspect of laser design that nature has not succeeded in producing by itself and that is the resonant cavity of a laser. Multiple passes through a laser amplifying medium improves the quality of laser operation immensely. This is achieved among humans using two mirrors but as yet this technological advancement seems to be missing in gas regions around stars.	5

Paragraph

Although a resonant cavity seems to be missing in natural masers there does seem to be a high degree of beaming. Small differences across the irregularly shaped maser cloud disk lead to vast differences in intensity due to exponential gain. The directions in the gas disk that have a longer length of population inversion will appear much brighter (as the increased maser amplification leads to an exponential increase). The majority of the radiation will emerge along this line of greatest length in a “beam”; this is termed *beaming*. Not quite as good as a laboratory laser cavity but pretty impressive for a seemingly random gas cloud.

Megamasers is the term used for water masers in the gas cloud around black holes.

Most large galaxies with a nucleus and a bulge have, at their centres, a supermassive black hole. When the black hole is actively accumulating matter, it releases a tremendous amount of energy, and the galaxy is said to have an active nucleus. Buffered by the dust in the flow of matter into the black hole, molecules can survive there and get energised by collisions with other molecules and dust particles. Water molecules are common in this environment, and they can emit maser radiation. A water maser can shine a beam of microwaves at a very specific frequency, 22.235 GHz, which corresponds to microwaves about 1 cm in wavelength. The primary reason megamasers have been studied in detail is that they make excellent tools to help us understand the environment around black holes. Small changes in the detected frequency of the masers, owing to the Doppler effect, mean that we can determine the line-of-sight velocity of maser clouds very precisely. All this means that the mass of black holes can be determined 20 times more accurately.

Another primary science goal that we can address with studies of water megamasers is measuring distances to galaxies. Measuring distances is a notoriously difficult problem in astronomy, and one of the most important. By analysing the internal dynamics of water maser systems, we can measure the rotation velocity, v , of maser clouds as they orbit the black hole from the Doppler shift of the maser lines. We can also measure the centripetal acceleration, a , of maser clouds by observing how the Doppler shift velocity changes over time. Using the simple relation for centripetal acceleration:

$$a = \frac{v^2}{r}$$

we can then calculate the radius of the gas ‘disk’ (see below) orbiting the black hole.

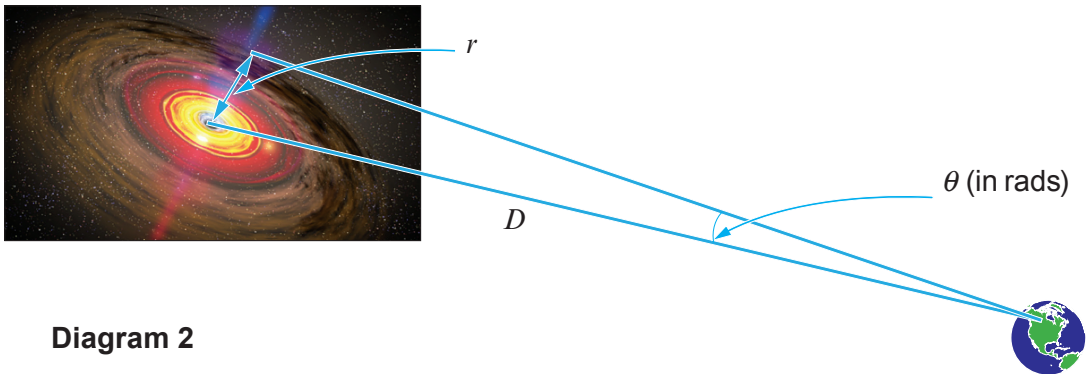


Diagram 2

We can also measure the angular size (θ) of the gas disk (in radians). Given the angular size (θ) and the radius, r , of the gas disk, we can obtain the distance, D by assuming that the angle (θ) is small. The simplicity of this method is remarkable and gives reliable results that are not dependent on controversial or unproven theories. In fact, the results obtained thus far from a couple of suitable galaxies with megamasers leads to a Hubble constant of $(68.9 \pm 7.1) \text{ km s}^{-1} \text{ Mpc}^{-1}$.

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7. Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

(a) In your own words explain what a megamaser is and how it works. (See paragraphs 1, 4, 7 and 8.) [5]

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(b) If the lifetime of a metastable level leading to visible light is $2\mu\text{s}$, estimate the lifetime of a 22.235 GHz maser transition. (See paragraph 2.) [3]

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- (c) The recessional velocity of the galaxy used in paragraph 10 is $3\,300\text{ km s}^{-1}$. Use Hubble’s law to calculate the distance of the galaxy in Mpc. (N.B. use the value of H_0 in paragraph 10. There is no need to change the units.) [2]

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- (d) What does the author mean by “We can also measure the centripetal acceleration, a , of maser clouds by observing how the Doppler shift velocity changes over time”? (See paragraph 9.) [3]

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- (e) (i) Doppler shift measurements are obtained for a gas disk orbiting a black hole. The orbital velocity of gas around the black hole is measured as 410 km s^{-1} and the acceleration of the gas disk is measured as 6 km s^{-1} **per year**. Calculate the distance of this region of the gas disk from the black hole. (See paragraph 9.) [3]

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- (ii) The angular size (θ) in radians of the gas disk is measured as 5.1×10^{-9} rad $\pm 10\%$ and its distance from Earth is measured as 1.53×10^{23} m $\pm 10\%$. Evaluate whether or not these values are consistent with your calculation in part (e)(i). (See Diagram 2.) [4]

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END OF PAPER

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Surname	Centre Number	Candidate Number
Other Names		2



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PHYSICS – A2 unit 3
Oscillations and Nuclei

MONDAY, 4 JUNE 2018 – AFTERNOON
2 hours 15 minutes

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator and a **Data Booklet**.

For Examiner's use only			
	Question	Maximum Mark	Mark Awarded
Section A	1.	9	
	2.	12	
	3.	8	
	4.	8	
	5.	8	
	6.	19	
	7.	16	
Section B	8.	20	
Total		100	

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space use the continuation page(s) at the back of the booklet taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

This paper is in 2 sections, **A** and **B**.
Section **A**: 80 marks. Answer **all** questions. You are advised to spend about 1 hour 35 minutes on this section.
Section **B**: 20 marks. Comprehension. You are advised to spend about 40 minutes on this section.
The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **4(b)**.

SECTION A

Answer all questions.

1. (a) A car travels at a constant speed of 45.0 km h^{-1} around a curve in the road with a radius of 80 m.
- (i) Explain why the car is accelerating. [2]
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- (ii) Calculate the angular velocity of the car (in rad s^{-1}) as it travels around the curve. [2]
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- (iii) Calculate the acceleration of the car and state its direction. [3]
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- (b) Discuss how the application of science enables cars to travel safely around curves. [2]
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2. Kinetic theory for an ideal gas gives an expression that can be written as:

$$pV = \frac{1}{3} nN_A m \overline{c^2}$$

(a) State the meaning of the terms: [2]

(i) m

(ii) $\overline{c^2}$

(b) Defining both symbols, explain what quantity is given by nN_A . [3]

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(c) The product of pressure and volume for an ideal gas may also be expressed as $pV = nRT$. Show in clear steps that the total translational kinetic energy of one mole of the gas is $\frac{3}{2} RT$. [4]

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- (d) An ideal monatomic gas, initially at a pressure of 115 kPa and temperature of 294 K, expands at constant pressure from a volume of $2.20 \times 10^{-3} \text{ m}^3$ to $2.60 \times 10^{-3} \text{ m}^3$. Calculate the change in the internal energy of the gas. [3]

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3. (a) Give the definition of the *specific heat capacity* of a material. [1]

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(b) An insulating flask contains $0.6 \times 10^{-3} \text{ m}^3$ of water at 19.5°C . A volume $1.6 \times 10^{-3} \text{ m}^3$ of boiling water at 100.0°C is poured into the flask.

(specific heat capacity of water, $c = 4\,200 \text{ J kg}^{-1}^\circ\text{C}^{-1}$; density of water, $\rho = 1\,000 \text{ kg m}^{-3}$)

(i) Determine the final temperature of the water in the flask. [3]

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(ii) Calculate the heat lost by the boiling water. [3]

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(iii) Justify the statement that no work is done on/by the boiling water as it cools. No further calculations are needed. [1]

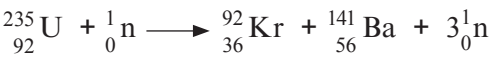
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4. (a) A fission nuclear reaction for uranium is:



The masses of the nuclei are:

mass of ${}^{235}_{92}\text{U}$ = 235.01 u

mass of ${}^{92}_{36}\text{Kr}$ = 91.90 u

mass of ${}^{141}_{56}\text{Ba}$ = 140.89 u

mass of ${}^1_0\text{n}$ = 1.01 u

1 u = 931 MeV

Calculate the energy released in this nuclear reaction.

[2]

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(b) Explain carefully the relevance of the **binding energy per nucleon** curve to nuclear fission and fusion. [6 QER]

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5. (a) Radon gas decays by emitting α -particles. It has a half-life of 3.8 days. Calculate the **percentage reduction** in the activity of a sample of radon after 12 days. [4]

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- (b) A student makes the following measurements for a radioactive source using the indicated absorber between the source and detector.

Absorber	Counts per minute
none	1 004
sheet of paper	597
2 mm of aluminium	23
15 cm of lead	27

Explain these observations. [4]

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6. (a) A load of 0.150 kg is attached to the lower end of a light vertical spring of stiffness (spring constant) 7.50 N m^{-1} . The system is initially in equilibrium. A student raises the load vertically through a distance of 0.095 m and releases it so that it oscillates.

Determine:

- (i) the extension of the spring when the system is in equilibrium; [2]

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- (ii) the period of oscillation. [2]

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- (b) The student notices the amplitude of oscillation decreasing and records the following amplitudes.

Oscillation number (n)	Amplitude (A) /m
0	0.095
10	0.062
20	0.043
30	0.029
40	0.019
50	0.014
60	0.009

She comments that “the rate of decrease of amplitude is larger at the start of the experiment than at its end”. Justify the statement using the data from the table. [2]

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(c) She expects that the measured amplitude of oscillation can be described by the equation:

$$A = A_0 e^{-\frac{n}{N}}$$

where A_0 is the initial amplitude, n is the oscillation number and N is a constant for the experiment that describes the decay of the amplitude.

(i) Justify that A_0 is the amplitude when $n = 0$. [1]

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(ii) By considering the amplitude when $n = 30$ use the equation to determine a value for N . [2]

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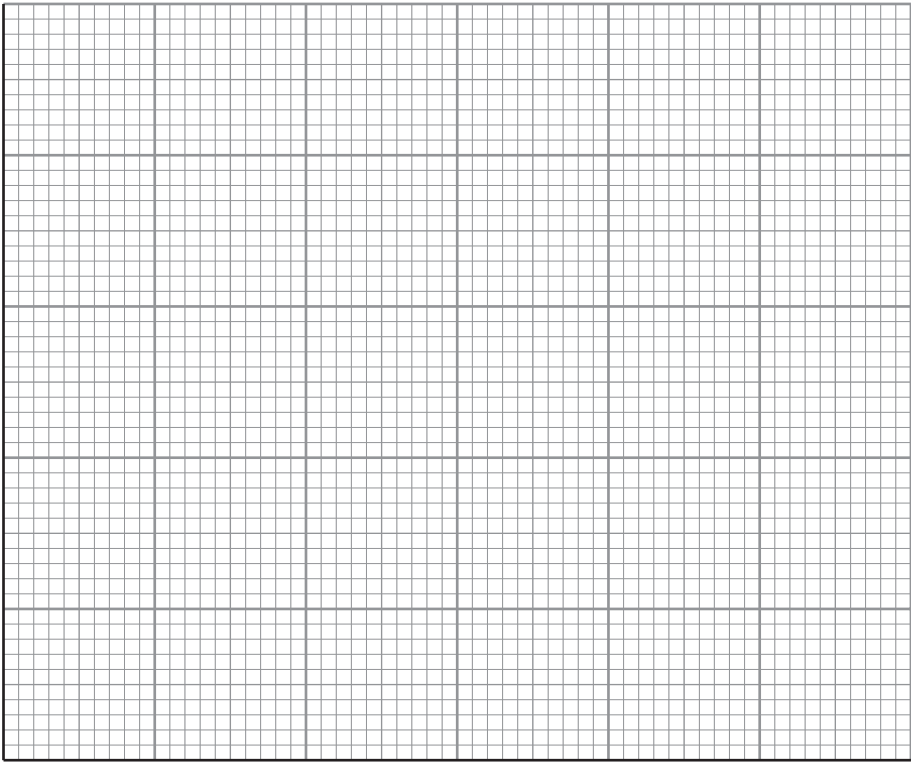
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(d) The student decides to check the validity of the equation.

(i) Complete the third column in the table. The first four rows have already been completed. [1]

Oscillation number (n)	Amplitude (A)/m	$\ln\left(\frac{A_0}{A}\right)$
0	0.095	0
10	0.062	0.43
20	0.043	0.79
30	0.029	1.19
40	0.019	
50	0.014	
60	0.009	

(ii) Plot $\ln\left(\frac{A_0}{A}\right)$ (y -axis) as a function of the oscillation number (x -axis) and draw a line of best fit. [3]



(iii) Explain whether or not your graph is in agreement with the equation $A = A_0 e^{-\frac{n}{N}}$. [2]

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(iv) Use the graph to determine a value for N . [3]

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(v) Explain which of the two values obtained for N is expected to be the more accurate, the value in part (c)(ii) or (d)(iv). [1]

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7. (a) A pendulum oscillating with simple harmonic motion consists of a small 0.05 kg mass oscillating at the end of a string of length 4.0 m. The displacement x in metres at time t can be written as:

$$x = 0.25 \cos \omega t$$

- (i) Show that the angular velocity, ω , is approximately 1.57 rad s^{-1} . [2]

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- (ii) Calculate the maximum speed of the mass. [2]

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- (iii) Show that the kinetic energy (E_k) of the mass, in joules, may be written as: [2]

$$E_k = 3.8 \times 10^{-3} \sin^2 1.57t$$

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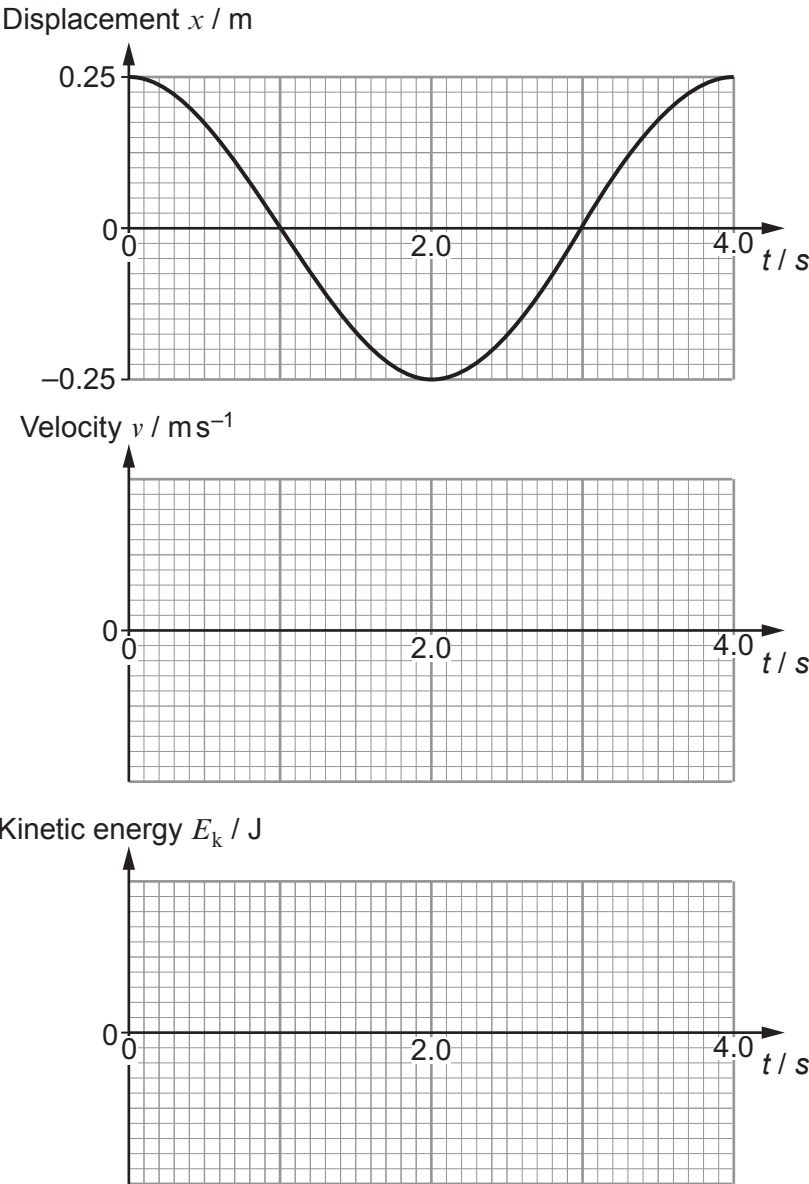
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- (iv) The displacement-time curve for the pendulum is shown for a time interval of one period i.e. 4.0 s. Draw curves for velocity, v , and E_k on the axes below for the same interval of time. Indicate values on your axes for v and E_k . [4]



- (v) Determine the maximum height reached by the pendulum mass above the level of its lowest point. [2]

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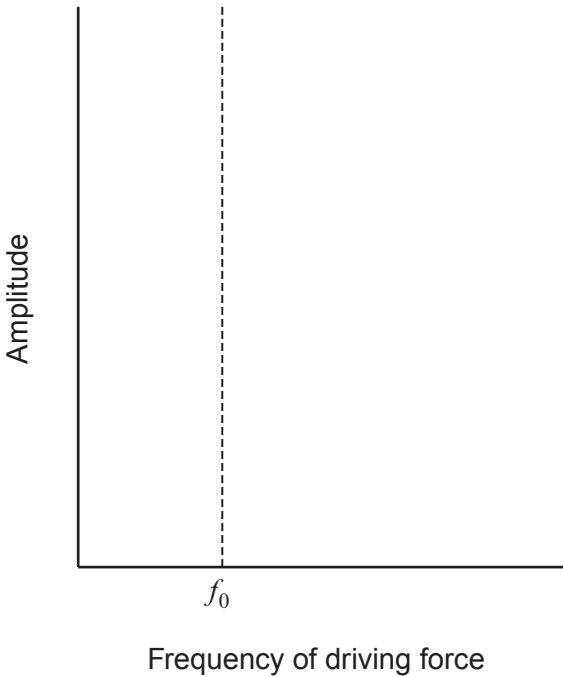
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Examiner
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(b) A system that can oscillate may be driven by an external sinusoidal force.

- (i) Sketch the amplitude of a lightly damped system as the frequency of the driver is increased. The natural frequency f_0 of the system is indicated. Label the curve **X**. [1]



- (ii) Name the effect seen near the frequency, f_0 . [1]

- (iii) Sketch a second curve on the **axes above** for an identical oscillating system that is damped to a greater extent. Label the curve as **Y**. [2]

SECTION B

Answer all questions

Read through the following article carefully.

Freely adapted from:

The Physics of Optical Fibres
By Justino Luis Moreno

Paragraph

Basically, an optical fibre is just a piece of glass along which you send light. A similar procedure was first demonstrated not with glass but with water flowing from a spout in 1840 by scientists Daniel Colladon and Jacques Babinet. Babinet later published his work in an article entitled “On the reflections of a ray of light inside a parabolic liquid stream”. This effect can be reproduced quite easily in a school lab using the apparatus shown in Figure 1. 1

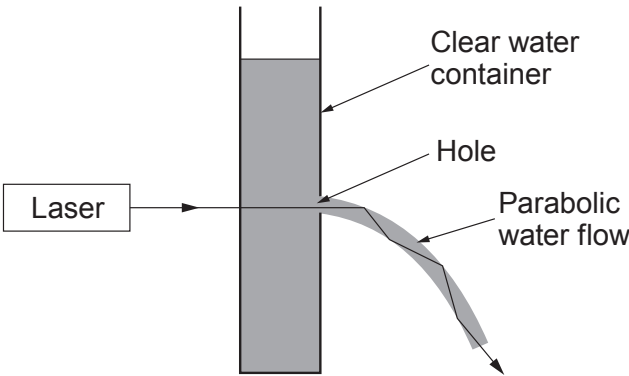


Figure 1

Although this set up is pleasing to the eye and is the basis of some water features, it isn't much use for international telecommunication! A standard optical fibre is shown in Figure 2 along with a ray of light entering it. 2

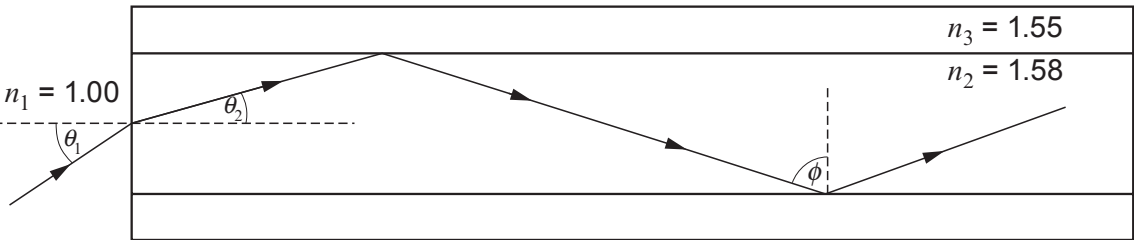


Figure 2

This ray of light is repeatedly reflected along the length of the optical fibre. If the entrance angle (θ_1) is small enough then an effect called total internal reflection (TIR) means that the light is completely reflected each time resulting in no light escaping as it travels along the optical fibre. The physicists in charge of designing these things can prove that the minimum value angle ϕ can have for TIR to take place is given by the equation: 3

$$\phi = \sin^{-1}\left(\frac{n_3}{n_2}\right)$$

This angle is around 80° for the optical fibre shown. The minimum angle for ϕ means that there is a maximum angle for θ_1 . This gives an acceptance cone where the input light gets propagated without loss. A quick bit of geometry shows you that the exit angle is the same as the entry angle so that when light exits it produces an almost perfect cone shaped beam with a circular cross-section as can be seen in Figure 3.

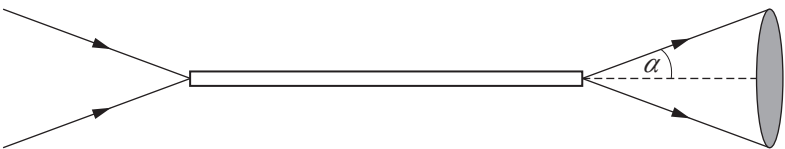


Figure 3

An important term in optical fibre technology is maximum bit rate. Since each pulse represents a bit of data, this is the highest frequency of pulses that can be sent down an optical fibre before the pulses start to overlap and become indistinguishable from each other. A monomode optical fibre cable of length 80km can comfortably have a maximum bit rate of 10 Gbs^{-1} . This means that 10^{10} pulses can pass along its length every second without overlapping. A typical telephone conversation requires around 10 kbs^{-1} meaning that one monomode optical fibre cable can carry a million telephone conversations simultaneously. High definition TV requires a much higher bit rate and a 10 Gbs^{-1} fibre will only handle around 2 000 high definition TV signals.

One of the most important factors that limit data transfer in optical fibres is multimode dispersion. This, put simply, is to do with the entrance cone of light in Figure 3. There is a range of distances that the pulses have to travel because there are a variety of angles at which they can travel along the fibre. The pulses then become spread out and indistinct, ruining the digital signal. Multimode dispersion is eliminated by using monomode optical fibre cables which have very thin cores. As a rule, monomode fibres have a core diameter of around $8\mu\text{m}$. This means that the core is less than 10 wavelengths thick so that the light stops behaving like rays. For monomode fibres there is only one propagation direction – along the axis.

Another drawback of sending signals down long lengths of optical fibres is that some of the light is either scattered or absorbed by the glass molecules themselves (an effect known as attenuation). Although no light escapes the fibre due to TIR there are other losses involved and these are usually summarised by using a decibel (dB) scale. This scale is defined by saying that a 10dB decrease (-10 dB) in power is when the power has dropped to 10% of its input value. A loss of -20dB then corresponds to a drop to 1% power and -30 dB is a drop to 0.1% power. The following table shows the relationship between dB values and power ratio:

dB	Power ratio $\left(\frac{P}{P_0}\right)$
-5	0.316
-10	0.100
-15	0.032
-20	0.010

Table 1

The losses of optical fibres are usually quoted in the unit dB/km and some modern optical fibres can have values as low as 0.01 dB/km. Hence, each km of cable loses 0.01 dB meaning that you can use 1 000km before your signal is down to 10% strength. Optical fibres have come a long way since they were born in a fountain of light nearly 200 years ago. They stretch out (literally) to all areas of the world bringing light, sound and broadband wherever they go. Technical advances mean that data can be sent at a rate of 1.05×10^{15} pulses per second over a distance of 50 km with only one monomode optical fibre. Nonetheless, the technology has its limitations of which attenuation and multimode dispersion are but two.

Examiner
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8. Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

(a) Explain why the water in Figure 1 flows in a downward curve (paragraph 1). [2]

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(b) Show that the minimum angle (ϕ) for total internal reflection in the optical fibre of Figure 2 is less than 80° (paragraph 3). [2]

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(c) Explain why the light emerging from the optical fibre is in the shape of a cone of light (paragraphs 3 & 4 and Figures 2 & 3). [2]

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(d) Calculate the angle (α) of the cone of light emerging from the optical fibre in Figure 3 (see paragraphs 3 & 4 and Figures 2 & 3). [4]

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(e) Rhys makes the following claim:

“A multimode optical fibre that can transfer data at a maximum bit rate of 500 Mb s^{-1} over a distance of 1 km would be able to transfer data at a maximum rate of 50 Mb s^{-1} over a distance of 10 km.”

Discuss whether or not Rhys’s claim is valid (paragraph 5 & 6). [3]

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(f) A certain type of optical fibre has an attenuation of 0.8 dB/km. When the signal decreases to 6% of its original intensity it must be amplified or the signal will be lost. An engineer intends to install these optical fibres in lengths of 20 km before each amplifier. Determine whether or not this is an appropriate length of optical fibre to use (see paragraph 7 and Table 1). [3]

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(g) Aled claims that a core diameter of around $13\mu\text{m}$ is thin enough for a monomode optical fibre because a wavelength in air of $1.3\mu\text{m}$ is usually used in communications. Rhian claims that this is nonsense because the wavelength in the optical fibre is changed due to its refractive index. Justify by including a calculation who is correct (paragraph 6). [4]

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[illegible]

Surname	Centre Number	Candidate Number
Other Names		2



GCE A LEVEL
1420U30-1
PHYSICS – A2 unit 3
Oscillations and Nuclei



THURSDAY, 6 JUNE 2019 – MORNING
2 hours 15 minutes

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator and a **Data Booklet**.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space use the continuation page(s) at the back of the booklet taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

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Section **A**: 80 marks. Answer **all** questions. You are advised to spend about 1 hour 35 minutes on this section.
Section **B**: 20 marks. Comprehension. You are advised to spend about 40 minutes on this section.
The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **4**.

For Examiner's use only			
	Question	Maximum Mark	Mark Awarded
Section A	1.	16	
	2.	15	
	3.	15	
	4.	6	
	5.	15	
	6.	13	
Section B	7.	20	
Total		100	

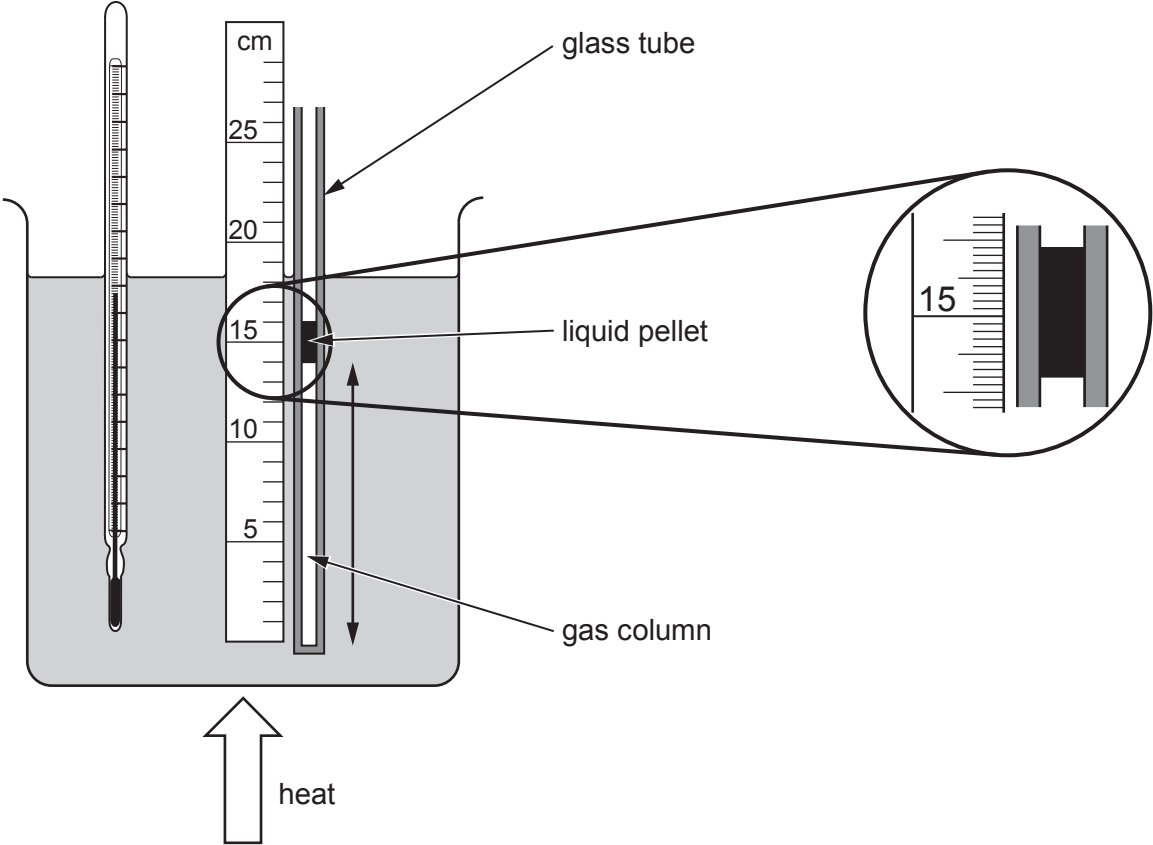


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SECTION A

Answer all questions.

1. A student performs an experiment to estimate the Celsius temperature of absolute zero.



The length, l , of the gas column and its temperature, θ , are measured at atmospheric pressure, 1.01×10^5 Pa. The temperature of the water is initially 0.0°C . It is then increased to 80.0°C . Readings are taken at 20.0°C intervals. The scale used to measure the length gives an uncertainty of ± 0.1 cm at the top of the column and an uncertainty of ± 0.1 cm at the bottom.

Values of θ and l are recorded in the table below.

$\theta / ^\circ\text{C}$	l / cm
0.0	11.5
20.0	12.5
40.0	13.2
60.0	14.2
80.0	15.0



Examiner
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(a) Justify the number of significant figures used to record the length, l . [2]

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(b) The glass tube in which the gas is trapped has an internal diameter of (1.5 ± 0.1) mm.

(i) Calculate the volume of the gas at 0.0°C . [2]

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(ii) Show that the percentage uncertainty in this volume, V , is approximately 15%. [3]

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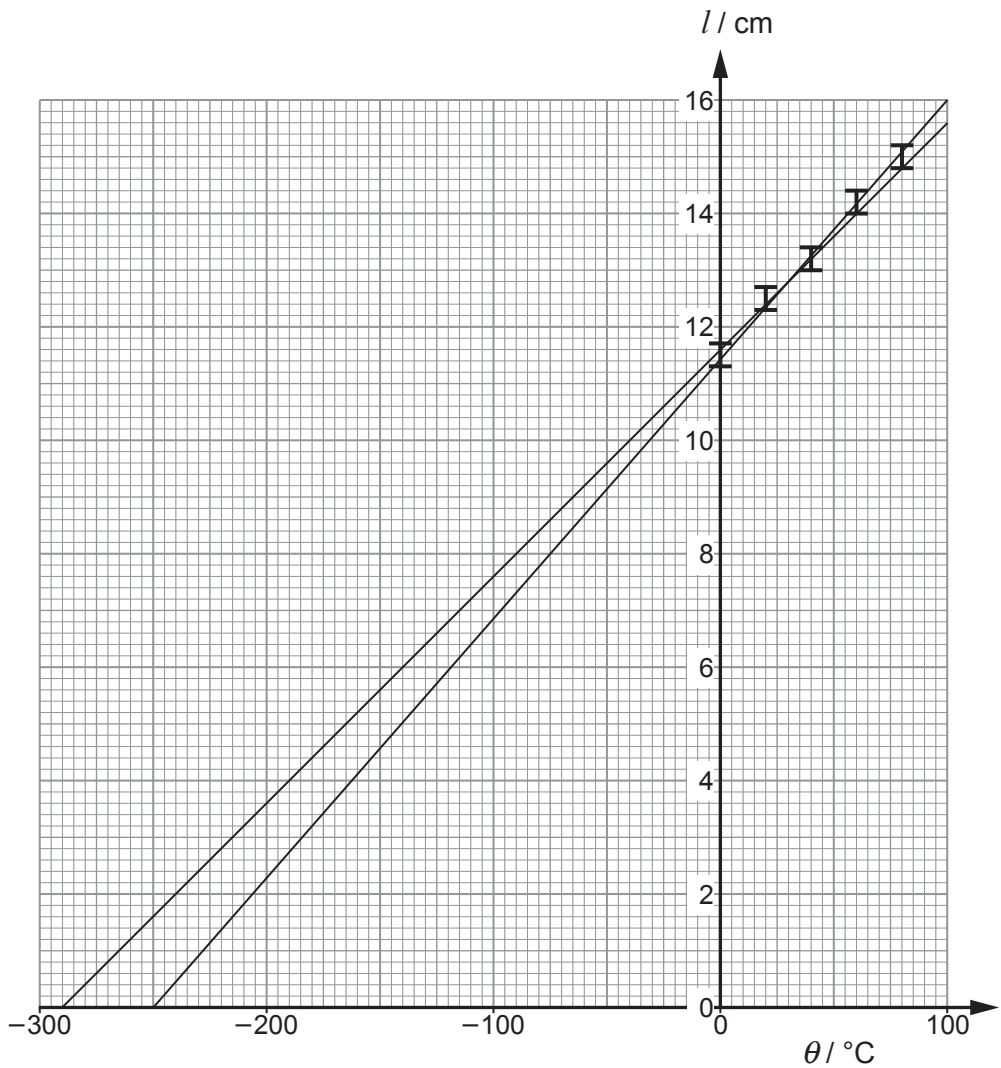
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- (c) The length, l , is plotted against the temperature, θ . Lines of maximum and minimum gradient consistent with the error bars have been drawn.



- (i) Use the graph to estimate the temperature of absolute zero in $^\circ\text{C}$, together with its **absolute** uncertainty. [2]

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(ii) Justify the statement that *the results are consistent with the ideal gas equation*. [4]

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(d) State what happens to the behaviour of the gas molecules as the temperature approaches absolute zero. [1]

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(e) The scale resolution of the ruler affects the final uncertainty in the value of absolute zero. Identify **two** other factors that could affect the accuracy. [2]

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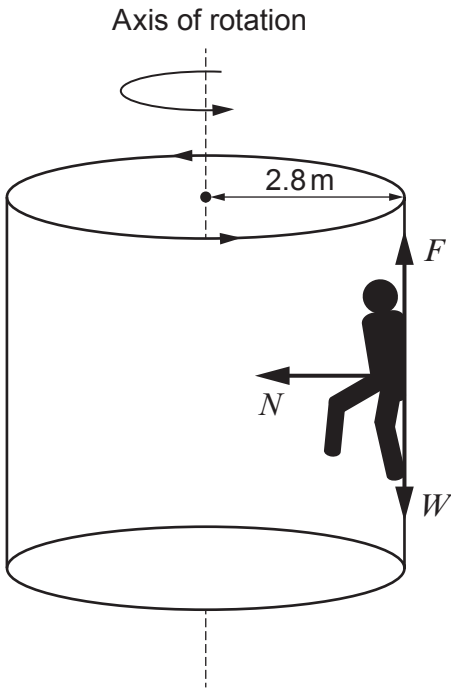
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2. A fairground ride consists of a large empty cylinder of radius 2.80 m with its axis of rotation vertical. A person of mass 66.2 kg walks into the cylinder and stands against its inner wall. The cylinder is rotated around the axis. When the rotation rate is 36.0 revolutions per minute the floor is lowered, but the person remains fixed in place against the wall.

This is a student's diagram.



- (a) Define the *radian*. [1]

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- (b) Calculate:
- (i) the period of rotation, T ; [2]

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- (ii) the speed, v , at which the person is moving in the circle. [2]

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- (c) Three forces act on the person: the normal contact force from the wall, N , the weight, W , of the person, and the frictional force, F , from the wall. These are shown in the student's diagram.

Explain why the values of N and F are approximately 2600 N and 650 N respectively. [4]

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- (d) The maximum value that a frictional force can take is μN where μ is a dimensionless constant.

- (i) Show that the person does not slide down the wall provided that μ is larger than approximately 0.25. [2]

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- (ii) A student says that as the cylinder slows down the floor will need to be raised to support the person when the angular velocity reduces below a certain value. Justify this and determine this angular velocity if μ has a value of 0.45. [4]

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3. Simple harmonic motion (SHM) may be represented by:

$$a = -\omega^2 x$$

(a) State what quantities are represented by: [1]

a

ω

x

(b) If the amplitude of the oscillation of an object moving with SHM is 0.012m and the period is 0.40s:

(i) determine the maximum acceleration; [3]

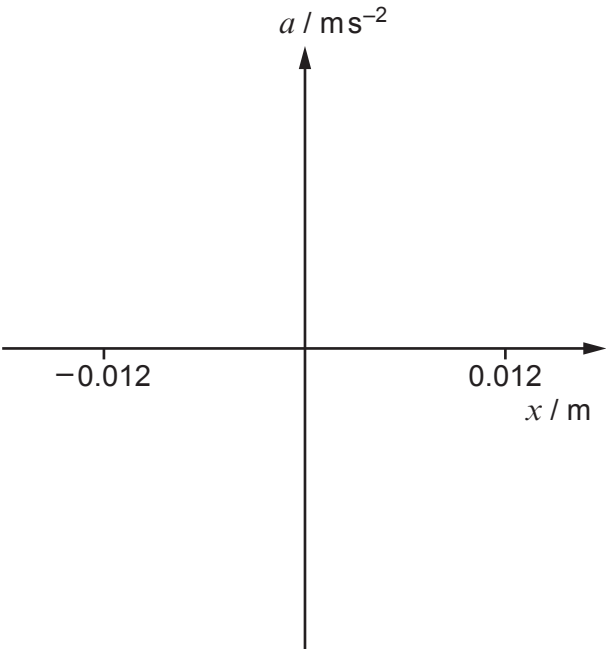
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(ii) sketch a graph showing the variation of a with x . Insert values on the a -axis. Values are given on the x -axis. [3]



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- (c) The oscillating object in part (b) is at its centre position when $t = 0$ and moving in the positive x direction. Insert the missing values in the expression for x in metres. [3]

$$x = \square \cos(\square t + \square)$$

- (d) An oscillating system may be driven by an external force. Describe and explain:

- (i) an application of forced oscillations that is useful; [2]

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- (ii) an example of forced oscillations that should be avoided. [3]

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4. Describe an experiment that uses a pendulum to determine the acceleration due to gravity with graphical analysis. [6 QER]

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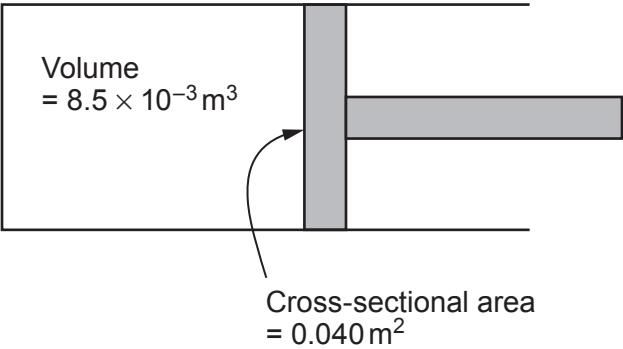
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5. A sealed container with a leakproof piston of cross-sectional area 0.040 m^2 at one end has an initial volume of $8.5 \times 10^{-3}\text{ m}^3$. It contains oxygen gas at a pressure of $5.0 \times 10^5\text{ Pa}$ and a temperature of 285 K .

(Relative molecular mass of oxygen = 32)



(a) Calculate:

- (i) the number of moles of oxygen gas in the container; [2]

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- (ii) the number of molecules in the container; [1]

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- (iii) the rms speed of the molecules; [3]

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(iv) the force exerted by the gas on the piston. [1]

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(b) The gas is allowed to expand at a constant temperature of 285 K to a volume of $10.2 \times 10^{-3} \text{ m}^3$.

(i) Calculate the final pressure of the gas. [2]

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(ii) During the process the gas does 773 J of work. Determine the heat flowing into the gas, explaining your reasoning. [2]

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- (iii) Following the expansion the gas is returned to its initial state by a two-stage process:
- a decrease in volume at constant pressure;
 - an increase in pressure at constant volume.

A student says that approximately 60 J of heat flows into the gas during the overall cycle (the expansion and return of the gas to its original state). Justify this. Include a sketch of the paths on a p – V diagram with values on both axes. [4]

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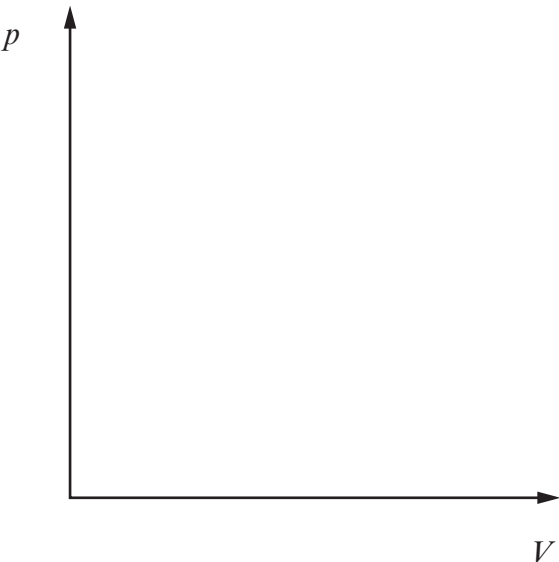
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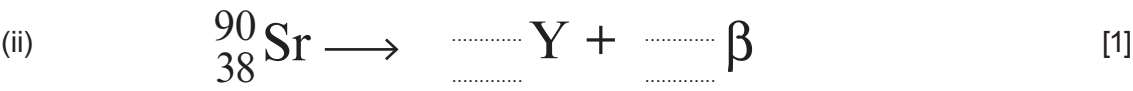
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6. (a) Complete the following decay equations.



(b) Calculate the binding energy **per nucleon** for ${}^{90}_{38}\text{Sr}$. [4]

$m_{\text{proton}} = 1.007\,276\,\text{u}$
 $m_{\text{neutron}} = 1.008\,664\,\text{u}$
 $m_{\text{electron}} = 0.000\,549\,\text{u}$
atomic mass of ${}^{90}_{38}\text{Sr} = 89.907\,738\,\text{u}$
 $1\,\text{u} = 931\,\text{MeV}$

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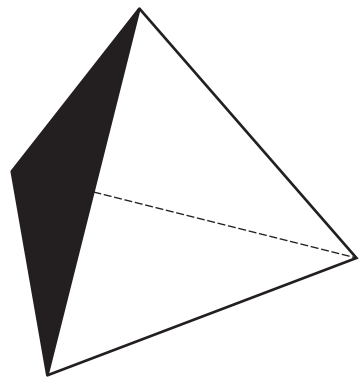
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(c) Radioactive decay may be investigated in the laboratory using a dice analogy.

A student has 564 identical wooden dice. These dice are tetrahedrons, with one face painted black. A tetrahedron is a solid with four identical faces, each face being an equilateral triangle.



(i) If the student throws one of the dice, what is the probability of the tetrahedron landing on the black face? [1]

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(ii) The student then throws all dice and counts and discards those that landed on the black face. The remaining dice are thrown again and the process is repeated multiple times. The number of dice discarded after each throw is recorded in the table, and the number remaining after each throw is calculated. The results are shown in the table, which also gives the fraction of dice remaining, where:

Fraction of dice remaining after a throw = $\frac{\text{number of dice remaining after a throw, } R}{\text{number of dice thrown in that throw, } T}$

Throw number, n	Number of dice thrown, T	Number of dice discarded after throw	Number of dice remaining after throw, R	Fraction remaining after throw, $\frac{R}{T}$
1	564	138	426	0.76
2	426	116	310	0.73
3	310	87	223	0.72
4	223	52	171	0.77
5	171	39	132	0.77
6	132	34	98	0.74
7	98	10	88	0.90
8	88	27	61	0.69
9	61	18	43	0.70
10	43	8	35	0.81



Examiner
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I. Explain why the expected fraction remaining after n throws is $(0.75)^n$. [2]

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II. Hence predict the expected number remaining after 10 throws. [1]

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III. Discuss to what extent the observations agree with those expected from theory. [3]

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SECTION B

Answer all questions

Read the following article carefully.

A Brief History of Particle PhysicsFreely adapted from <http://www.particleadventure.org/other/history/quantum.html>

Paragraph

The story starts a long time ago (~400BC) in ancient Greece. Democritus, Leucippus and Epicurus developed the theory that *the universe consists of empty space and an (almost) infinite number of invisible particles* which differ from each other in *form, position, and arrangement*. All matter is made of indivisible particles called atoms.

1

It could be argued that very little happened for another 2000 years until in 1654 Otto von Guericke invented a vacuum pump. You will probably have heard about one use he made of this vacuum pump – the Magdeburg hemispheres. Two metal hemispheres with a vacuum between them could not be separated by teams of horses pulling them apart.

2



Otto von Guericke also invented an electrostatic generator – electron accelerators were born and lightning discharge could be produced artificially.

3

In 1705, it was noted that by combining both inventions of Otto von Guericke, lightning discharge can go between electrodes in a gas and can go further at low pressures.

4

In 1838, Michael Faraday noted a strange glow from low pressure gases in glass tubes when the gas was conducting electricity. Some 40 years later Eugen Goldstein noted so-called “cathode rays” in these gases. In 1898, Joseph (JJ) Thomson identified these cathode rays as streams of negatively charged particles and made measurements on the properties of these electrons. He then put forth his “plum-pudding” model of the atom. In this model, the atom is a slightly positive sphere with small, raisin-like negative electrons inside – he obviously didn’t know that radioactivity, researched by Marie Curie, was a nuclear phenomenon. Hardly surprising since the nucleus hadn’t been discovered yet....

5

The early 20th century saw Hans Geiger and Ernest Marsden, under the supervision of Ernest Rutherford, scatter 4.7 MeV alpha particles off a gold foil and observe large angles of scattering. This put an end to the plum-pudding model and suggested that atoms have a small, dense, positively charged nucleus of diameter around 10^{-14} m surrounded by electrons orbiting at a distance of around 10^{-10} m.

6



Paragraph

Shortly afterwards, Niels Bohr succeeded in constructing a theory of atomic structure based on quantum physics. Then, in 1924, Louis de Broglie proposed that matter had wave-like properties.

7

Strangely enough, de Broglie’s wave particle duality led to JJ Thomson’s son (George Paget Thomson) obtaining one of the very first electron diffraction patterns for crystals. Whereas his dad obtained a Nobel prize for identifying the electron as a particle, he obtained the prize for its wave properties.

8

The first evidence for a proton was found by Ernest Rutherford in 1919 and, in 1921, James Chadwick and E.S. Bieler concluded that some strong force holds the nucleus together.

9

Paul Dirac realised that the positively charged particles required by his equation were new objects (he called them “positrons”). They are exactly like electrons, but positively charged – the first example of antiparticles. James Chadwick discovered the neutron. Both these developments took place in 1931.

10

Enrico Fermi put forth a theory of beta decay that introduced the weak interaction. This is the first theory to explicitly use neutrinos and particle flavour changes (1933-34).

11

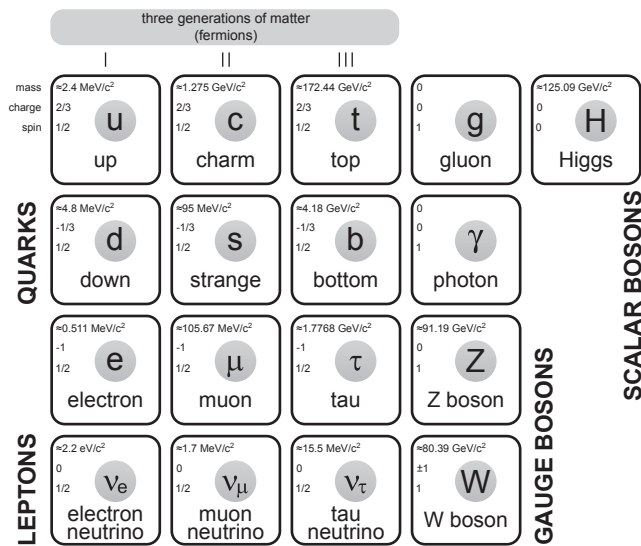
In 1947, a meson that interacts strongly was found in cosmic rays, and was determined to be the pion. A year later, the Berkeley synchro-cyclotron produced the first artificial pions.

12

Some 20 years later, in the late 1960s, electron scattering experiments were carried out using the Stanford Linear Accelerator. High speed electrons were scattered off protons and the electrons appeared to be bouncing off multiple small hard cores inside the proton. James Bjorken and Richard Feynman analysed this data in terms of a model of constituent particles inside the proton.

13

Standard Model of Elemental Particles



In a summary talk for a conference, John Iliopoulos presented, for the first time in a single report, the view of physics now called the Standard Model – this was 1974.

14

After eighteen years of searching with many accelerators, experiments at the Fermilab near Chicago discovered the top quark at the unexpected mass of 172 GeV. No one understands why the mass is so different from the other five quarks.

15

Almost half a century after Peter Higgs predicted a Higgs boson as part of a mechanism (invented by a number of theorists) by which fundamental particles gain mass, the ATLAS and CMS experiments at the CERN lab discovered the Higgs boson (2012).

16

Some two and a half millennia after the commencement of particle physics it seems we are a lot closer to understanding the Universe but there are still many questions to be answered. Some of the most fundamental questions to be answered are: What is dark matter? What is dark energy and do either of these exist? How does quantum gravity work? For the answers to all these questions, and any others, please consult your physics teacher.

17



Examiner
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7. Answer the following questions in your own words. Extended quotes from the original article will not be awarded marks.

(a) Calculate the maximum force required to separate two Magdeburg hemispheres of radius 10 cm (atmospheric pressure = 1.01×10^5 Pa, see paragraph 2). [2]

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(b) Suggest why lightning discharge travels further at low pressures (see paragraphs 4 & 5). [2]

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(c) (i) Calculate the distance of nearest approach for a 4.7 MeV alpha particle when fired at gold foil (the charge on a gold nucleus is $+79e$, see paragraph 6). [3]

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(ii) State why your answer to (c)(i) and the results of the alpha particle scattering experiment contradict the “plum-pudding” model (see paragraphs 5 & 6). [2]

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Examiner
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- (d) A student claims that a 4 keV electron is suitable for atomic diffraction experiments. Determine whether or not she is correct (see paragraphs 6 & 8). [4]

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- (e) Explain why the strong nuclear force is required to hold nuclei together (see paragraphs 6 & 9). [1]

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- (f) A typical thermal electron has a rms speed of approximately $100\,000\text{ ms}^{-1}$. When a positron annihilates an electron at room temperature, two photons of wavelength $2.43 \times 10^{-12}\text{ m}$ are produced. Explain, using conservation of momentum, why these photons travel in opposite directions (see paragraph 10). [3]

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(g) Explain why the quark make up of a π^- meson is $\bar{u}d$. [1]

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(h) Calculate the mass of a top quark in kg (see paragraph 15).
Note that $1\text{ u} \equiv 931\text{ MeV}$ if required. [2]

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END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		2



GCE A LEVEL

1420U30-1



THURSDAY, 26 MAY 2022 – AFTERNOON

PHYSICS – A2 unit 3
Oscillations and Nuclei

2 hours 15 minutes

For Examiner's use only			
		Question	Maximum Mark
Section A	1.	13	
	2.	14	
	3.	19	
	4.	13	
	5.	21	
Section B	6.	20	
Total		100	

ADDITIONAL MATERIALS

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The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **3(b)**.



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SECTION A

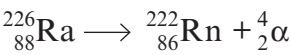
Answer **all** questions.

1. (a) Radon is a radioactive gas and emits alpha particles.
- (i) Explain what is meant by the *activity* of a radioactive source **and** give its unit. [2]
- (ii) The half-life of radon is 3.3×10^5 s. Determine the time it takes for a sample of radon to decay to 20 % of its initial activity. [3]
- (iii) Discuss the extent to which alpha particles are a risk to the human body. [3]



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(b) Radium-226 decays into radon.



Assuming that 98 % of the energy released in this process is converted to the kinetic energy of the alpha particle, use the data below to determine the speed of the alpha particle. [5]

Atomic masses: ${}^{226}_{88}\text{Ra} = 226.025\,410\,\text{u}$ ${}^{222}_{86}\text{Rn} = 222.017\,578\,\text{u}$

${}^4_2\text{He} = 4.002\,603\,\text{u}$ $m_{\text{electron}} = 0.000\,549\,\text{u}$

$1\,\text{u} = 931\,\text{MeV}$

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2. A balloon of volume $1.60 \times 10^{-3} \text{ m}^3$ contains helium at a pressure of $1.20 \times 10^5 \text{ Pa}$ and a temperature of 293 K . (Relative molecular mass of helium = 4.)

(a) Determine:

(i) the number of molecules in the balloon; [2]

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(ii) the rms speed of the molecules. [2]

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(b) The product of the pressure and the volume of the gas is given by $pV = nRT$ and also by $pV = \frac{1}{3}Nmc^2$.

(i) Show that the mean kinetic energy of a gas molecule is $\frac{3}{2}kT$ where k is the Boltzmann constant. [4]

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(ii) Calculate the mean kinetic energy of a molecule in the gas. [1]

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(iii) Determine the internal energy of the gas in the balloon. [1]

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(c) (i) Explain, in terms of the molecules, the difference in the nature of the **internal energy** in an ideal gas and in a liquid. [3]

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(ii) State the temperature of a system for its internal energy to be a minimum. [1]

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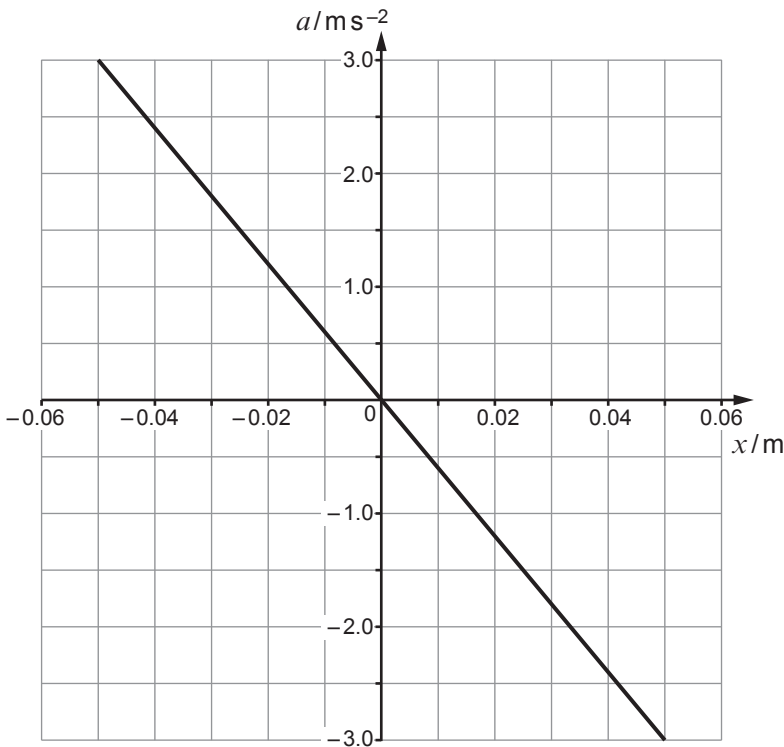
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3. A small object of mass 0.30 kg oscillates at the end of a spring. The graph shows how its acceleration, a , depends on its displacement, x , from a fixed point.



(a) (i) Identify the **two** characteristics **of the graph** that show that the object is oscillating with *simple harmonic motion*. [2]

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(ii) Use the graph to show that the angular frequency, ω , of the oscillation is approximately 7.7 rad s^{-1} . [3]

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(iii) Calculate:

I. the period of oscillation; [1]

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II. the spring constant; [2]

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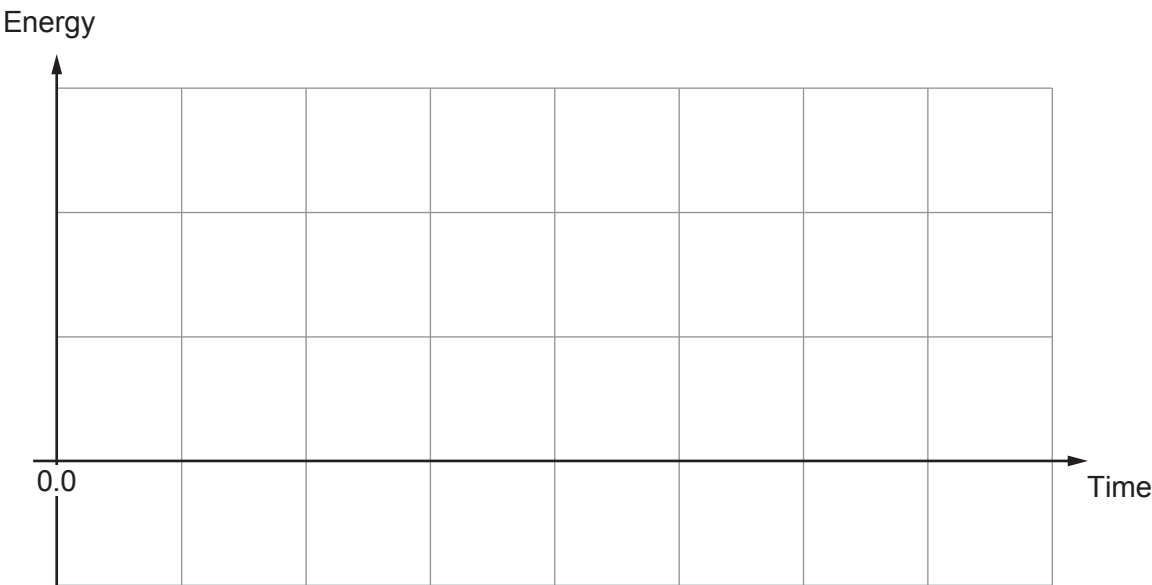
III. the velocity of the object as it passes through the centre of oscillation. [2]

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(iv) The object was released at the displacement of -0.05 m at time $t = 0$. Sketch on the axes below:

- I. the potential energy of the system (label the sketch PE),
- II. the kinetic energy of the system (label the sketch KE).

Draw both sketches **over one cycle** of the undamped oscillation. (No values are required on the axes.) [3]



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(b) Explain what is meant by the terms *damping* and *resonance* **and** discuss their importance in real systems giving an example for each.

[6 QER]

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4. A satellite orbiting the Earth completes one revolution in 105 minutes.

Mass of the Earth = 6.0×10^{24} kg, radius of the Earth = 6400 km

(a) Explain what is meant by the term *radian*. [2]

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(b) Calculate the satellite's angular velocity, ω , in rad s^{-1} . [3]

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(c) The gravitational force on the satellite is given by $\frac{GM_E m}{R^2}$ where m is the mass of the satellite, M_E is the mass of the Earth and R is the radius of the orbit. Show that:

$$R = \sqrt[3]{\frac{GM_E}{\omega^2}}$$
 [2]

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Examiner
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(d) Determine the height of the satellite above the surface of the Earth. [3]

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(e) The mass of the Moon is 7.3×10^{22} kg and its radius is 1 740 km. Discuss whether a satellite would be able to orbit the Moon with a period of 105 minutes. [3]

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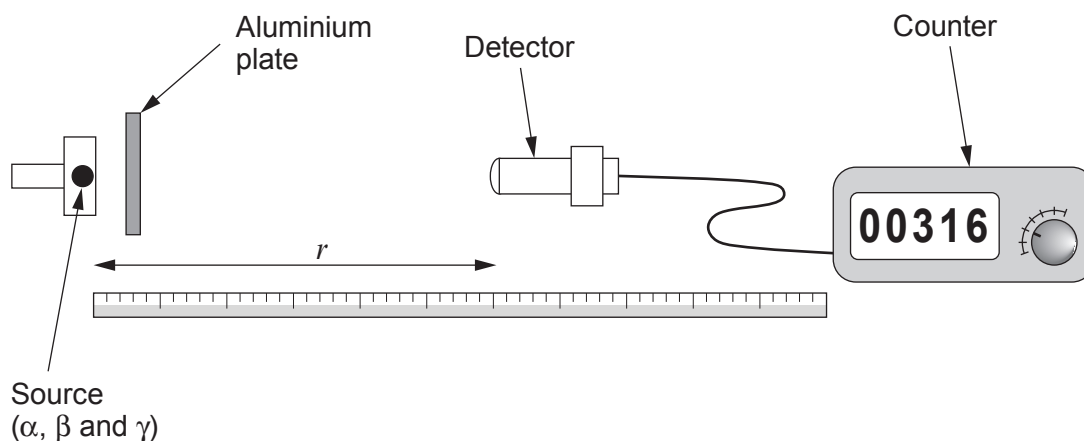


5. A student says that the radiation intensity, I , of gamma rays increases with decreasing distance, r , from a source such that:

$$I \propto \frac{1}{r^n}$$

She wishes to determine the value of n .

The student uses a source of alpha, beta and gamma radiation. She measures the intensity of the rays at set distances from the source by determining the count over intervals of one minute. She also takes measurements without the source at the start and at the end of the experiment.



- (a) (i) Suggest why a thin aluminium plate is placed near the source. [1]

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- (ii) Explain why readings are taken without the source, at the start **and** at the end of the experiment. [2]

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- (b) At each distance, r , the counts were measured over one minute intervals three times. The uncertainty in r is ± 0.005 m. The measurements are recorded in the table.

Complete the table.
Space for calculations.

[2]

Distance, r / m	Count				Uncertainty in mean count
	First reading	Second reading	Third reading	Mean	
without source	19	22	25	22	3
1.000	110	130	136	125	13
0.800	195	165	178		
0.600	270	316	300		
0.400	661	604	651	639	29
without source	20	20	25	22	3

- (c) (i) The student subtracts the mean count without the source from the mean count for each distance to obtain the corrected mean count, N . She also determines the uncertainty in each of these corrected mean counts.

Complete the table.
Space for calculations.

[2]

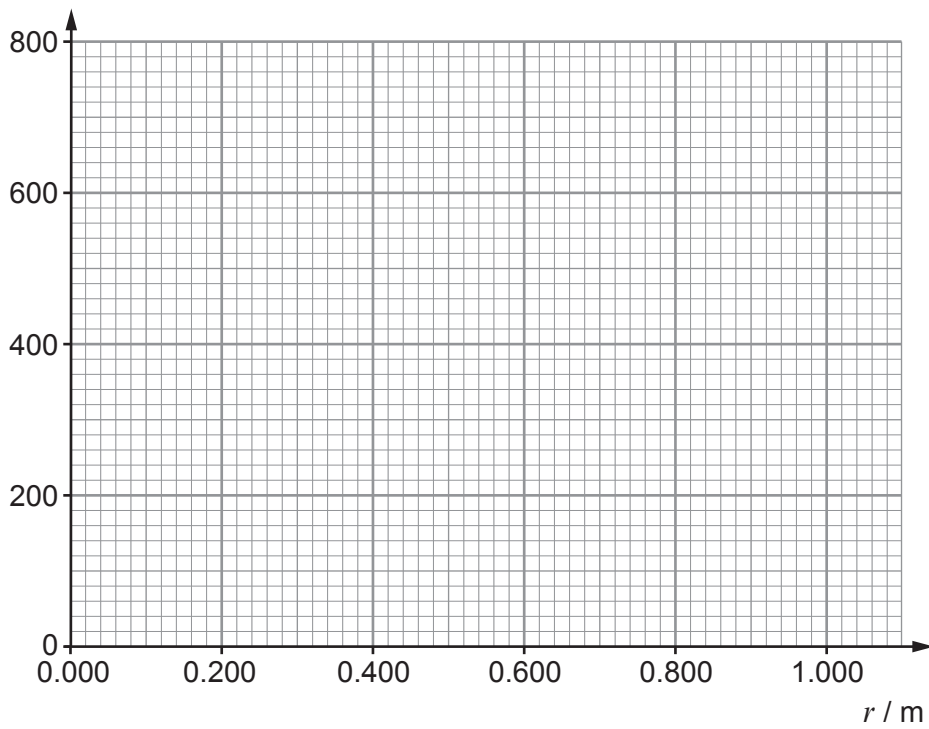
Distance, r / m	Corrected mean count, N	Uncertainty in corrected mean count
1.000	103	16
0.800		
0.600		
0.400		



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- (ii) Plot a graph of the corrected mean count, N , against distance, r . **Error bars are not required.** [3]

Corrected mean count, N



- (iii) The student claims that error bars may be shown on the graph for the corrected mean count, but not the distance. Justify her claim. Calculations are not required. [1]

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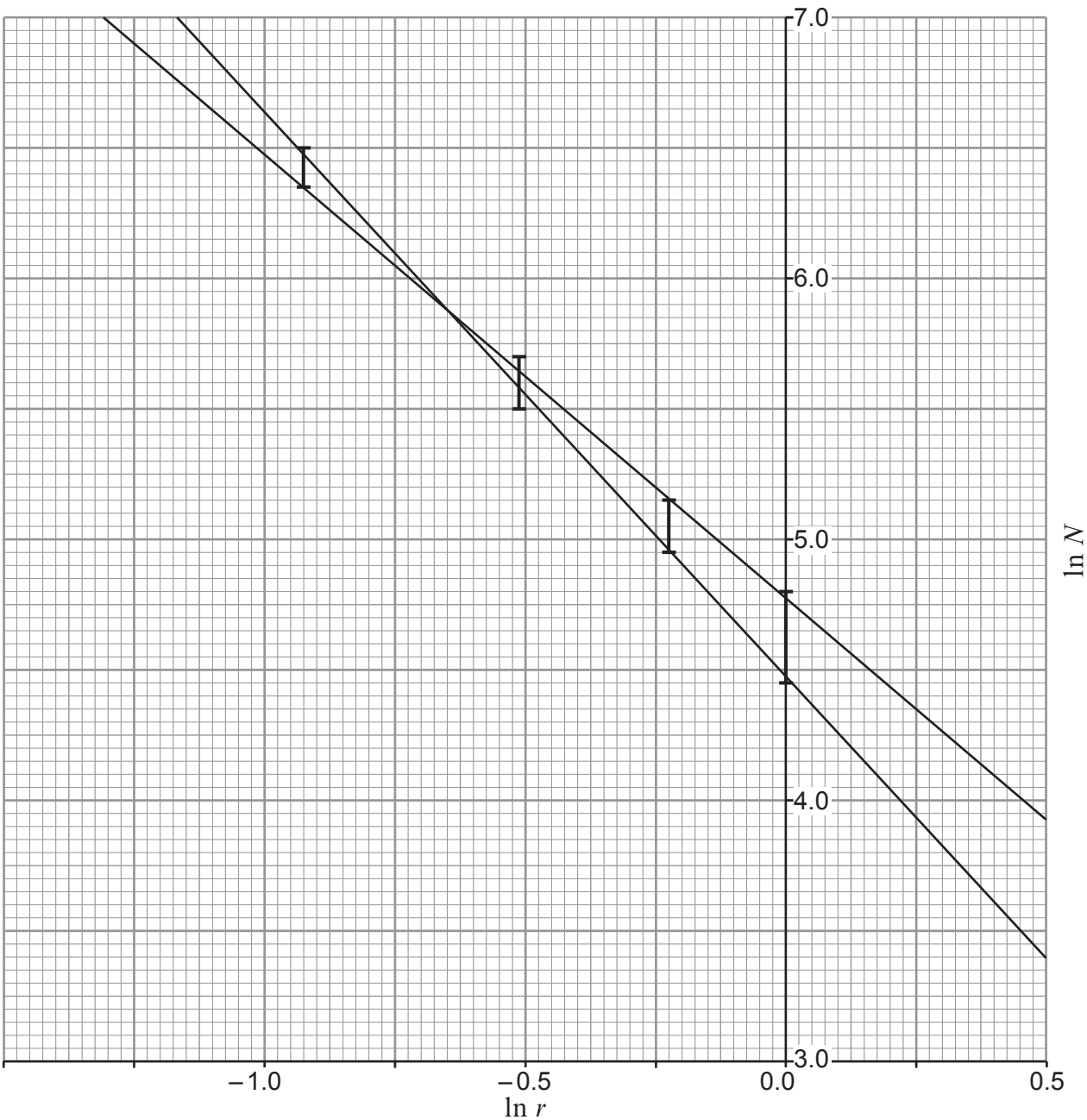
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- (d) Her friend suggests using a logarithmic graph. She calculates the logarithmic values in the table below.

Distance, r / m	$\ln r$	$\ln N$
1.000	0.00	4.63
0.800	-0.22	5.06
0.600	-0.51	5.61
0.400	-0.92	6.42

The data are plotted on the graph.



Examiner
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For a distance of $r = 1.000\text{ m}$ use the table in part (c) to show that the uncertainty in the logarithmic counts for this distance is approximately 0.15. [3]

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(e) (i) Starting with intensity: $I \propto \frac{1}{r^n}$ justify how the logarithmic graph can be used to determine n . [3]

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(ii) Use the maximum and minimum gradients to determine n and the **absolute** uncertainty in its value. [4]

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SECTION B

Answer **all** questions.

6. Read through the following article carefully.

How do you produce really low temperatures and can you then do some cool stuff with it?

Paragraph

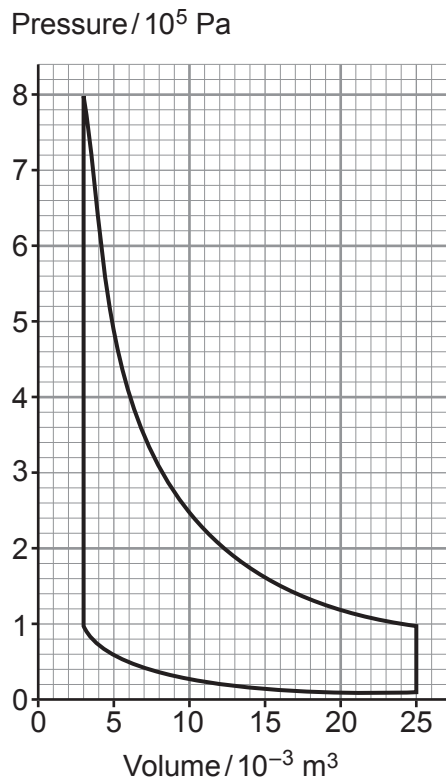


Figure 1

The first thing you have to do is to produce low temperatures and this is usually achieved by taking a gas through a closed cycle such as that shown in Figure 1. 1

1. Air, initially under high pressure, is expanded at constant temperature. 1
2. Heat is allowed to escape at constant volume. 2
3. The gas is compressed at a constant low temperature.
4. The gas is heated at constant volume to a high pressure.

The end result is that the gas does a large amount of work in each cycle. This means that the gas must have heat flowing into it leading to the cooling of whatever it is that requires cooling. 3

A different and simpler method of cooling gases is called the Joule-Thompson method, which does not involve a gas going through a cycle. A gas is compressed to a pressure of two thousand times atmospheric pressure. The gas is then allowed to cool back to room temperature. After that, it is allowed to expand quickly through a nozzle to atmospheric pressure leading to a tremendous amount of further cooling. In the expansion, the gas does a very large amount of work but the non-ideal gas molecules also have an enormous increase in potential energy because of the attractive forces between molecules. Both these factors cool the gas by a huge factor. 4

Okay, so now we have the basic physics for producing low temperatures. Apply this low temperature to air and we can make a product that can give hours of fun – liquid nitrogen (LN₂). 5

Most of the fun that can be obtained from LN₂ is based on the fact that it boils at a temperature of –195.8 °C and, in doing so, expands by a factor of approximately 1 000. For example, a cool rotating rocket can be built using a simple plastic water bottle. 6



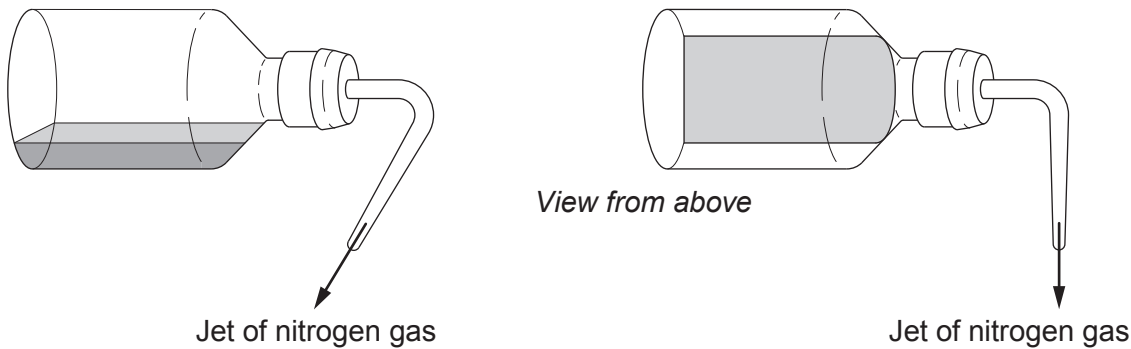


Figure 2

Paragraph

The boiling nitrogen makes the nozzle of the plastic water bottle into a jet engine. When you then place the plastic water bottle on a nicely polished wooden floor it will rotate far more quickly than you would ever anticipate, providing a great demonstration for even the most ardent physics-hating pupil. What can be even more spectacular is when the bottle accidentally hits something solid (like the leg of a lab desk). The wall of the plastic water bottle will be at around -200°C and fragile. This, combined with the high pressure inside the bottle, leads to rather a loud and impressive bang.

Although LN_2 can be fun, it can also be dangerous. For instance, when transporting a large, open container of LN_2 (see Figure 3), you should not remain in the confined space of a lift with the container in case the lift becomes stuck. Most of the danger from LN_2 arises from its 200°C temperature difference from its surroundings. Nonetheless, it is always amusing when TV chefs wear safety goggles and gloves when making ice cream using LN_2 but never bother with any safety equipment when using litres of boiling fat above a gas fire. I promise you that the boiling fat is far more dangerous even though the temperature difference, compared with human body temperature, is around 200°C for both LN_2 and boiling fat.

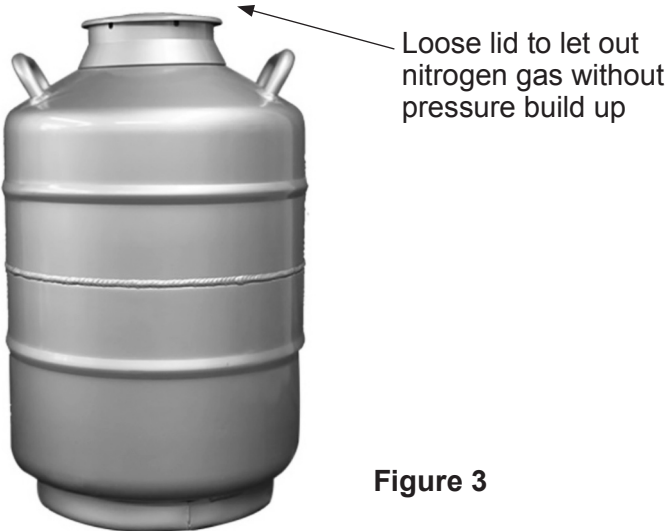


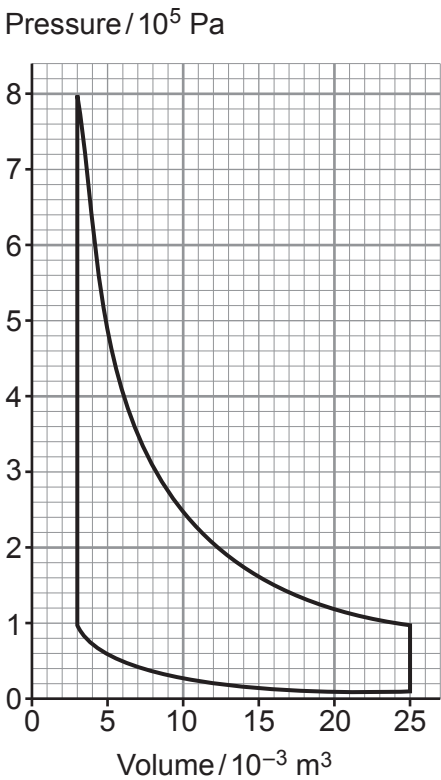
Figure 3

In conclusion, low temperatures and LN_2 can be dangerous for some obvious reasons but they can also lead to some amusement and even some Michelin-starred ice cream.



Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

- (a) **Label the diagram** with the processes 1–4 along with an arrow to show the direction of the cycle (see paragraph 2). [2]



- (b) The author states that work is done by the gas in the cycle. By considering the four stages of the cycle, explain whether or not this statement is correct (see paragraph 3 and Figure 1). [2]

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(c) Estimate the work done during the cycle. [3]

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(d) When a gas is compressed in a metal cylinder it will become hot. Explain why the gas will then cool to room temperature (see paragraph 4). [2]

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(e) Explain whether or not the author is correct to state that the gas expanding quickly through the nozzle will cool greatly (see paragraph 4). [4]

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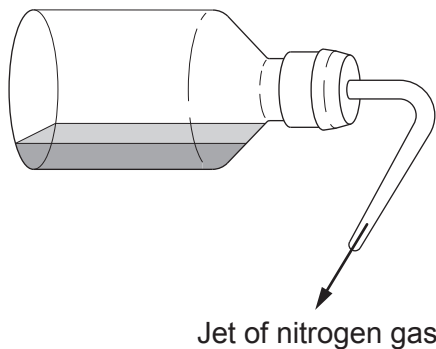
(f) Explain very briefly why nitrogen gas escapes through the nozzle of the plastic water bottle (see Figure 2 and paragraph 7). [1]

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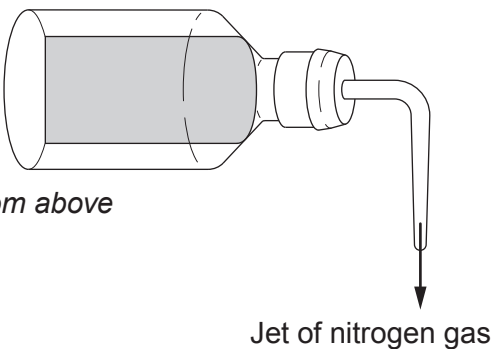
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- (g) Starting from Newton's 3rd Law, explain why the plastic water bottle shown will rotate anti-clockwise. [2]



View from above



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- (h) An exploding bottle produces a “loud bang” (see paragraph 7). Explain briefly why this “loud bang” can be heard more than 100 m from the exploding bottle. [2]

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- (i) Justify the author’s statement that it would be dangerous to share a confined space (like a lift) with a large container of liquid nitrogen (see paragraph 8 and Figure 3). [2]

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END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		2



GCE A LEVEL

1420U30-1



FRIDAY, 9 JUNE 2023 – MORNING

PHYSICS – A2 unit 3
Oscillations and Nuclei

2 hours 15 minutes

ADDITIONAL MATERIALS

In addition to this examination paper, you will require a calculator and a **Data Booklet**.

For Examiner's use only			
	Question	Maximum Mark	Mark Awarded
Section A	1.	10	
	2.	15	
	3.	14	
	4.	9	
	5.	10	
	6.	22	
Section B	7.	20	
Total		100	

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

This paper is in 2 sections, **A** and **B**.
Section **A**: 80 marks. Answer **all** questions. You are advised to spend about 1 hour 35 minutes on this section.
Section **B**: 20 marks. Comprehension. You are advised to spend about 40 minutes on this section.
The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **3(d)**.



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SECTION A

Answer **all** questions.

1. The first law of thermodynamics may be written as:

$$\Delta U = Q - W$$

(a) State the meaning of: [3]

ΔU

Q

W

(b) A closed cylinder fitted with a leak-proof piston as one end contains 0.060 mol of an ideal monatomic gas. The volume of the system is $1.4 \times 10^{-3} \text{ m}^3$ and the pressure of the gas is 100 kPa. Determine the temperature of the gas. [2]

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(c) The gas is heated and expands at constant pressure to a volume of $2.0 \times 10^{-3} \text{ m}^3$. Determine:

(i) the work done by the gas, [2]

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(ii) the increase in the internal energy of the gas, [2]

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(iii) the heat transferred to the gas. [1]

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2. (a) Explain what is meant by centripetal force and state its direction. [2]

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(b) A small sphere of mass 30 g at the end of a light string rotates in a horizontal circle of radius 0.80 m and completes 10 revolutions in 15 s.

(i) Show that the speed of the sphere is 3.35 m s^{-1} . [2]

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(ii) Calculate the centripetal force on the sphere. [2]

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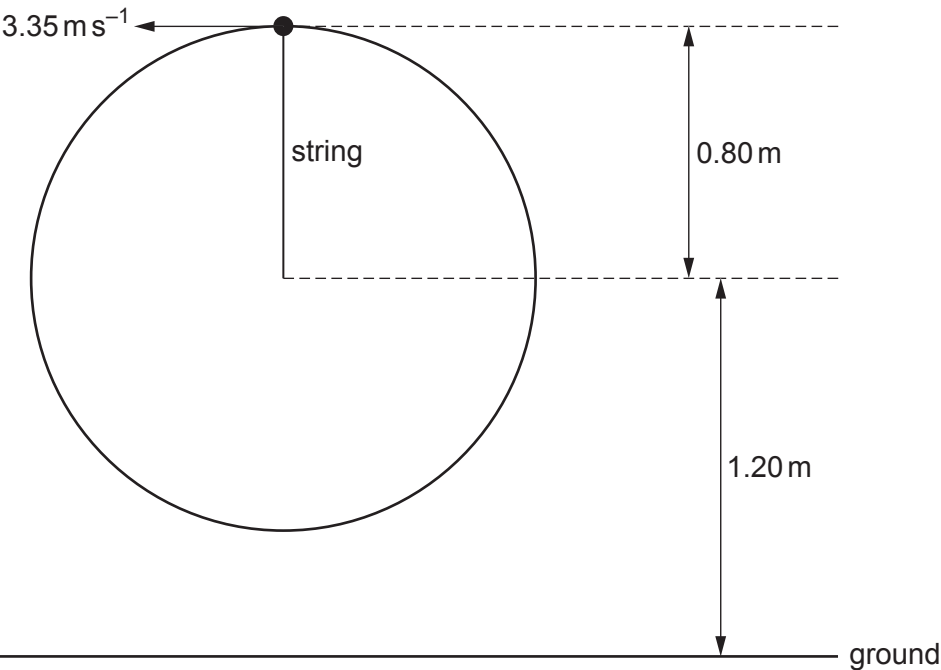
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- (c) The sphere now rotates in a **vertical** circle of the same radius, from a point that is 1.20 m above the ground. The speed at the top of the circle is 3.35 m s^{-1} .



Calculate the tension in the string at the **top of the circle**. [2]

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- (d) (i) Calculate the speed of the sphere when it reaches the **bottom of the circle**. [3]

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(ii) A student claims that if the string breaks when the sphere is at the **top of the circle** it will reach the ground at a horizontal distance of approximately 2 m away from the point of release. Investigate if her claim is correct, justifying your answer. [4]

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3. (a) State what is meant by alpha emission. [2]

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(b) Americium-241 has a half-life of 432 years. Determine the percentage **decrease** in the activity of the element after 30 years. [4]

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(c) A builder is asked to install a smoke detector. He researches smoke detectors and finds that the radioactive source in the detector emits alpha particles. He decides that it is too dangerous to install the detector. Discuss whether or not he has used science well in his decision making. [2]

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(d) Describe how alpha, beta and gamma radiation can be distinguished:

- by using different absorbers
- by using an electric field **or** a magnetic field.

[6 QER]

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4. (a) Explain what is meant by specific heat capacity. [2]

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(b) A volume of $1.2 \times 10^{-3} \text{ m}^3$ of water is heated in a 3 kW kettle.

Specific heat capacity of water (and tea) = $4\,200 \text{ J kg}^{-1} \text{ K}^{-1}$

Specific heat capacity of milk = $3\,900 \text{ J kg}^{-1} \text{ K}^{-1}$

Density of water = $1.00 \times 10^3 \text{ kg m}^{-3}$

Density of milk = $1.03 \times 10^3 \text{ kg m}^{-3}$

(i) Determine the time it takes to increase the temperature of the water from 18°C to 100°C . [3]

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(ii) A volume of $0.25 \times 10^{-3} \text{ m}^3$ of the heated water is used to make tea and this is poured into a thermos flask. The temperature of the tea when in the flask is 95°C . Milk at 5°C is then poured into the flask. Determine whether $3.6 \times 10^{-5} \text{ m}^3$ of milk will lower the temperature of the mixture to the required temperature of 84°C . [4]

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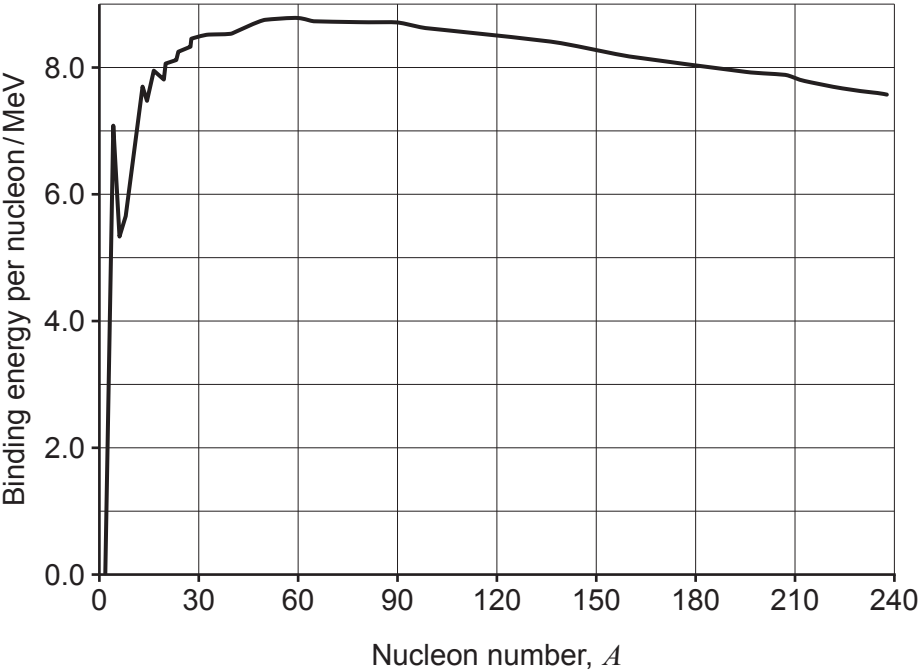
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5. (a) Nuclear energy can be released by fusion and fission. Discuss these processes in terms of energy and the stability of the nuclei. Use the graph of the binding energy per nucleon in your answer. [4]



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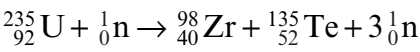
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(b) One of the many possible fission reactions of uranium-235 is:



The masses are:

${}^{235}_{92}\text{U}$: 235.04393 u; ${}^{98}_{40}\text{Zr}$: 97.91273 u; ${}^{135}_{52}\text{Te}$: 134.91645 u, ${}^1_0\text{n}$: 1.00866 u
and 1 u = 931 MeV.

(i) State what is meant by the Avogadro constant. [1]

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(ii) Calculate the energy released by the fission of 0.0010 kg of uranium-235 in this reaction. Give your answer in J. [5]

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6. A student says that a simple pendulum oscillates with simple harmonic motion with a period, T , of:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where l is the length of the pendulum and g is the acceleration due to gravity.

- (a) Describe what is meant by a simple pendulum. [1]

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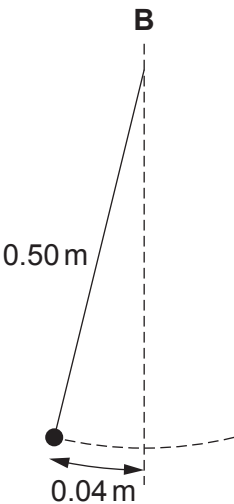
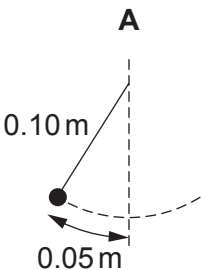
- (b) State what is meant by simple harmonic motion. [2]

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- (c) The student sets up two experimental systems to investigate simple harmonic motion. System A is a pendulum of length 0.10 m oscillating with an amplitude of 0.05 m. System B is a pendulum of length 0.50 m oscillating with an amplitude of 0.04 m. Give two reasons why system B is better than system A to investigate simple harmonic motion. [2]



Diagrams not to scale.

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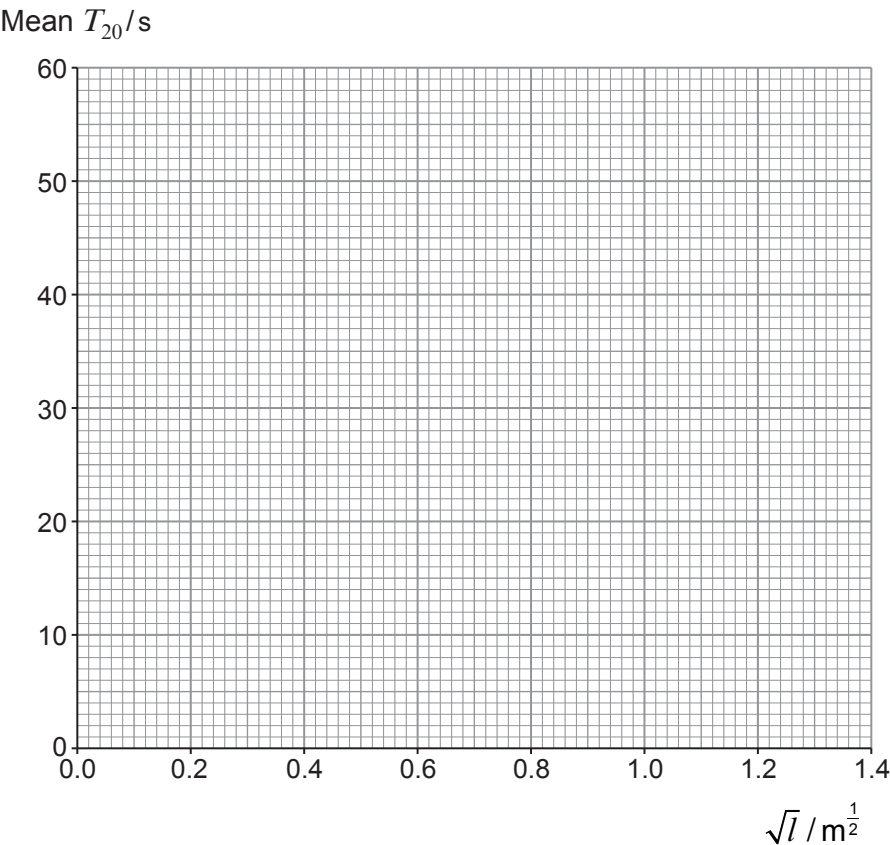
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- (d) Measurements for an oscillating simple pendulum are given in the table below, where l is the length of the pendulum and T_{20} is the time taken for 20 oscillations. For each value of l four values of T_{20} are measured.

l / m	$\sqrt{l} / \text{m}^{\frac{1}{2}}$	T_{20} / s	Mean T_{20} / s	Uncertainty T_{20} / s
0.250	0.50	19, 21, 22, 20	21	2
0.500	0.71	26, 29, 26, 29	28	2
0.750		35, 34, 33, 36		
1.000		42, 39, 40, 41		
1.250		45, 42, 44, 45		
1.500		48, 50, 49, 52		

- (i) **Complete the table** by determining $\sqrt{l}$, the mean time for 20 oscillations, T_{20} and the uncertainty in T_{20} . Values have been inserted for the first two lengths. [3]
- (ii) Plot mean T_{20} against $\sqrt{l}$ on the grid below. Include error bars on the T_{20} axis only. Draw a line of maximum gradient and a line of minimum gradient. [4]



Examiner
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(iii) Justify why error bars for $\sqrt{t}$ are not plotted on the graph. [2]

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(iv) Use the maximum and minimum gradients to determine a mean value for the acceleration due to gravity, g , and its **percentage** uncertainty. [5]

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(e) Another student says that the gradient of the graph would be approximately twice as large if the experiment was carried out on the Moon, where the acceleration is approximately $0.2g$. Investigate if this statement is correct, justifying your answer. [3]

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SECTION B

Answer **all** questions.

7. Read through the following article carefully. Paragraph

A summary of binary stars

First of all, we call two stars orbiting their common centre of mass a “binary star”. Binary stars fall into 5 different classes, categorised by the detection method: 1

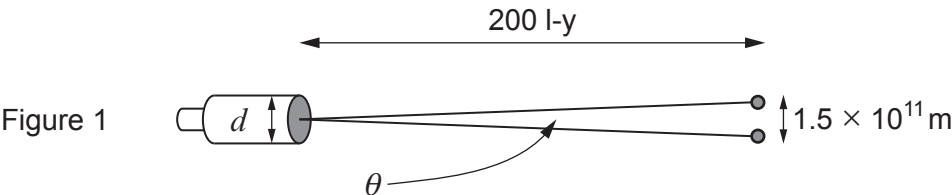
- 1. Visual binaries
- 2. Spectroscopic binaries
- 3. Eclipsing binaries
- 4. Astrometric binaries
- 5. Non-eclipsing binaries that can be detected using photometry.

Visual binaries

These do what it says on the tin – the angular separation of the stars is great enough that both stars can be seen as separate objects through a telescope. There is a simple relationship between the minimum angular separation θ (in radians), the diameter, d , of the telescope and the wavelength, λ , of the radiation used. 2

$$\theta = 1.22 \frac{\lambda}{d} \quad \text{Equation 1}$$

This equation means that a nearby binary star that is around 200 light years (l-y) away will only be visible as two stars with a $d = 8.0\text{ m}$ telescope if the two stars are separated by at least the Earth-Sun distance of $1.5 \times 10^{11}\text{ m}$ (see Figure 1). 3



Perhaps the most famous of visual binary stars is the brightest star in the night sky – Sirius. Sirius is not, in fact, a single star but a binary consisting of Sirius A and Sirius B that orbit each other with a period of around 50 years. Sirius A is a main sequence star which has roughly 2 times the mass of the Sun, 1.7 times the Sun’s temperature and 25 times its luminosity. Sirius B is a white dwarf with a similar mass to our Sun, a surface temperature 4.3 times greater than our Sun but a luminosity 18 times less than our Sun. 4

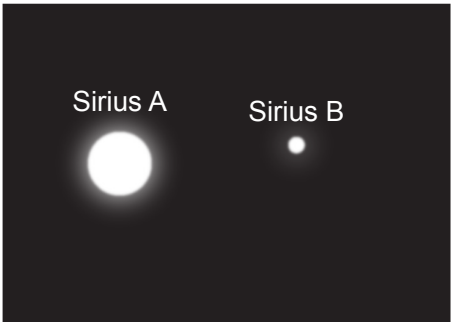


Figure 2

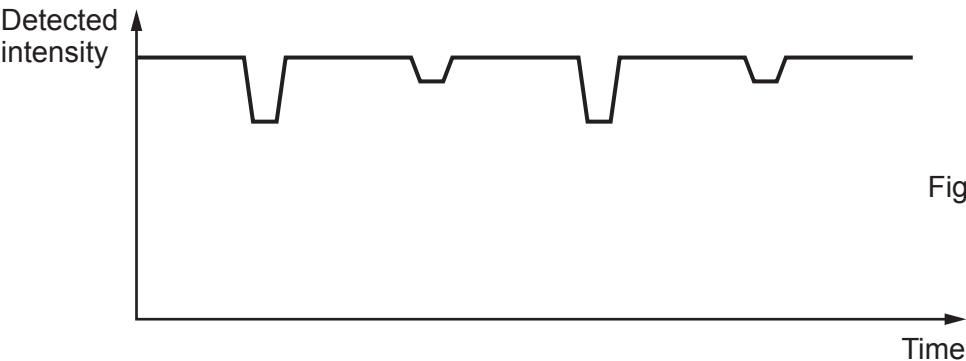


Spectroscopic binaries

These are binaries where the stars cannot be viewed separately but the existence of the two stars can be inferred from the variation of the Doppler shifted light from the system. Both stars will orbit their mutual centre of mass and hence their Doppler shifts will vary periodically during their orbits. This theory is the same as that used to discover exoplanets from the wobble of their parent star. 5

Eclipsing binaries

These happen when the line of sight of the observer is in the plane of the orbit of the two stars. When this happens, one star can pass directly in front of the other leading to a drop in the intensity of light entering the observer’s telescope. An idealised graph of intensity against time is shown in Figure 3. 6



In general, there are two eclipses for each orbit. The larger of the two dips in intensity occurs when the hotter of the two stars is blocked by the colder star. 7

Astrometric binaries

These are binary systems where only one star is visible and that star can be seen moving in an orbit around something which is invisible. These are different from spectroscopic binaries because the visible star can actually be seen to move rather than inferring its motion from the variation of the Doppler shift. Stars orbiting neutron stars have been observed by this method but the ultimate goal of this technique would be to observe a star orbiting a black hole. 8

Non-eclipsing binaries that can be detected using photometry

This simply means detecting binaries by measuring light intensity but not via the eclipsing method described earlier. It turns out that there are three different techniques of finding binaries within this class of binary. First, an increased light intensity can be detected periodically from the binary system – this is caused by light being reflected from one star, off the other and into the Earth’s telescope. Second, one star can exert such a strong gravitational pull on the other that it becomes deformed – this leads to a periodic fluctuation in the light intensity detected. Third, is an effect sometimes known as Doppler beaming – a star moving towards an observer will lead to an increased number of photons detected per second (as well as a decrease in the wavelength). 9

These 5 different classes of binary stars can provide an enormous amount of valuable data regarding stellar orbits and masses but they can also allow measurement of galactic distances and even give insights into black holes and supernovae. 10



Examiner
only

Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

- (a) A light year is 9.46×10^{15} m. Use equation 1 and a wavelength of 500 nm to show that the author’s claim about an 8 m telescope viewing the binary star mentioned in paragraph 3 and Figure 1 is correct. [4]

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- (b) (i) Show that the distance between Sirius A and Sirius B is approximately 3×10^{12} m (see paragraph 4). [Mass of the Sun is 2×10^{30} kg.] [2]

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- (ii) Show that the centre of mass of the Sirius binary system is located approximately 1×10^{12} m from Sirius A. [2]

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Examiner
only

- (iii) Use data in paragraph 4 and Stefan’s law to calculate the ratio of the radii of Sirius A and Sirius B:

[4]

$$\frac{R_{\text{Sirius A}}}{R_{\text{Sirius B}}}$$

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- (c) (i) Evaluate whether the variation of light intensity shown in Figure 3 is consistent with an eclipsing binary system (see paragraph 6).

[3]

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- (ii) Use Stefan’s law to explain why the “larger of the two dips in intensity occurs when the hotter of the two stars is blocked by the colder star” (see paragraphs 6 and 7 and Figure 3).

[2]

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Examiner
only

(d) Explain why a star orbiting a much more massive black hole is an astrometric binary (see paragraph 8). [1]

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(e) One of the three methods for detecting non-eclipsing binaries that can be detected by photometry, suggests that stars do not behave as ideal black bodies. Explain which **one** of the three methods relies on stars not behaving as black bodies (see paragraph 9). [2]

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END OF PAPER

